Semiparametric SDF Estimators for Pooled, Non-Traded Cash Flows

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Declaration of interest

The authors report no conflict of interest. The authors alone are responsible for the content and writing of the paper.

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Abstract

We develop a semiparametric framework for estimating stochastic discount factors from pooled non-traded cash flows. The framework generalizes the inception-based approach of Driessen et al. (2012), clarifies its relation to Korteweg and Nagel (2016), and accommodates dependence induced by overlapping funds. We show that horizon choice creates a finite-sample bias-variance-like tradeoff: averaging pricing errors over additional compounding dates can substantially improve estimator stability, even though longer horizons alter the population pricing restriction. We apply the framework to private equity and discuss its implications for identification and model specification in empirical work with sparse, illiquid cash-flow data.

1 Introduction

Private equity (PE) has grown into a major asset class, with more than $13 trillion in assets under management (McKinsey & Company, 2024). Yet rigorous asset-pricing tools for private markets remain comparatively underdeveloped. The central difficulty is that private equity investments do not generate reliable traded-return series. Instead, the econometrician observes irregular capital calls, distributions, and reported net asset values (NAVs), often over long horizons and under substantial informational frictions. As a result, standard return-based asset-pricing methods that are routine in public markets are difficult to apply in private markets, where the relevant pricing restrictions are expressed in discounted cash flows rather than in observed returns.

Semiparametric stochastic discount factor (SDF) methods provide a natural response to this measurement problem. The seminal contribution of Driessen et al. (2012) shows how to estimate abnormal performance and systematic risk exposure from pooled private-equity cash flows by imposing a zero-net-present-value restriction at fund inception. Korteweg and Nagel (2016) develop a closely related framework that links private-equity cash flows to public-market benchmarks while emphasizing the role of fund overlap and cross-sectional dependence. These approaches have become central tools in empirical work on private assets because they avoid the strong distributional assumptions required by fully parametric models and can be implemented directly on cash-flow data.

At the same time, existing semiparametric approaches leave open an important practical question: how do cash-flow-based SDF estimators behave in the finite samples researchers actually observe? Private-equity panels are typically sparse, nonlinear, and heavily overlapping across funds and vintages. In addition, recent vintages often contain large reported NAV components, so empirical conclusions may reflect continuation values and sample construction as much as underlying risk. In such settings, conclusions about alpha, market exposure, or multifactor structure may depend not only on economics, but also on horizon choice, estimator variance, and the treatment of cross-fund dependence.

This paper studies semiparametric cash-flow-based SDF estimation when the econometrician is not restricted to discounting all pooled cash flows back to fund inception. Our starting point is simple. In typical private-equity samples, inception-only pricing errors can be highly unstable because the estimating equations are nonlinear and the effective cross-sectional sample size is limited. A natural response is to average pricing errors over additional compounding dates rather than evaluate all discounted cash flows only at inception. But this is not merely a statistical refinement. Once pricing errors are evaluated at horizons beyond inception, the estimator generally targets a different population object. Horizon choice therefore creates a tradeoff between finite-sample stabilization and population target shift: longer horizons can improve finite-sample behavior by smoothing the sample criterion, but they also change the underlying population pricing restriction through what we term a timing-risk-premium component.

The paper's first contribution is conceptual. We derive a generalized pooled-cash-flow pricing framework that makes explicit how the choice of discounting date affects the estimand. This formulation nests the inception-only criterion of Driessen et al. (2012) as a special case. It also clarifies the relation to Korteweg and Nagel (2016): their estimator is closely connected to our framework, but it is not obtained as a literal special case because the order of aggregation differs. Framing the problem in this way makes explicit a distinction that, to our knowledge, has not been formalized in the semiparametric private-equity SDF literature: a horizon may improve finite-sample performance without preserving the original inception-based pricing restriction.

Our second contribution is methodological. We formulate the estimator as a nonlinear least-mean-distance (LMD) problem for pooled, non-traded cash flows (Pötscher and Prucha, 1997). This objective-function representation is useful because it makes the role of horizon choice transparent, accommodates alternative weighting schemes, and yields a clean comparison to existing semiparametric estimators. Building on this representation, we develop an increasing-domain asymptotic framework for inference under dependence induced by overlapping funds and adjacent vintages. In private-equity applications, cross-sectional units are neither independent nor naturally ordered in calendar time alone; dependence is tied to economic proximity, such as vintage-year distance and overlap in cash-flow histories. Our asymptotic results provide a large-sample benchmark for that setting. In the empirical application, however, we treat finite-sample simulation evidence and stability diagnostics as the primary basis for interpretation whenever asymptotic approximations appear unreliable.

The paper's central tension has two distinct margins. On one margin, inception-only discounting preserves the original zero-net-present-value pricing restriction but can produce unstable estimates in realistic private-equity samples. On the other margin, averaging pricing errors over additional compounding dates can materially improve finite-sample performance, but it does so by changing the population pricing restriction. Our goal is not to treat horizon choice as a purely mechanical implementation detail, but to characterize this tradeoff explicitly and provide guidance on how researchers should navigate it in empirical work.

We study this problem in three steps. First, we derive the generalized pooled-

cash-flow objective and the corresponding asymptotic framework. Second, we use simulations calibrated to private-equity-like cash-flow environments to evaluate how horizon choice affects finite-sample bias and variance. Third, we implement the estimator in the Preqin cash-flow database for North American U.S.-dollar buyout (BO) and venture-capital (VC) funds and assess specification stability using both asymptotic inference and hv-block cross-validation diagnostics (Racine, 2000) tailored to overlapping cash-flow dependence.

Our empirical results yield three main messages, all of which we interpret as diagnostics about identification under currently available private-equity cash-flow panels rather than as definitive statements about the population factor structure of private equity. First, including recent vintages with large NAV components can mechanically inflate measured market exposure, underscoring the importance of sample construction in private-equity asset pricing. Second, within our sample, parsimonious single-factor market specifications applied to seasoned vintages are substantially more stable than richer alternatives. Third, α terms and additional factors are only weakly identified in the available data and become highly sensitive to Horizon choice and finite-sample noise, even when asymptotic inference is supplemented with cross-validation diagnostics. Taken together, these results suggest that, in our sample and under our semiparametric design, current cash-flow panels support more stable inference about broad market exposure in seasoned buyout and venture-capital portfolios than about abnormal performance or richer factor decompositions.

These findings matter for more than estimator design. A large literature asks whether private-equity funds outperform public benchmarks and how their risks should be measured (comprehensively summarized by Korteweg (2019, 2023)). Our results suggest that these questions cannot be separated from the identification limits imposed by sparse, overlapping, and partially NAV-based cash-flow panels. In particular, our evidence does not imply that private-equity alpha is zero, nor does it invalidate prior estimates of outperformance as such. Rather, it indicates that semiparametric alpha estimates from pooled cash-flow data can be weakly identified and highly sensitive to implementation choices in samples of the kind currently available. That qualification is important for interpreting both positive and negative findings in the broader literature on private-equity outperformance.

The paper contributes to the literature on private-market performance measurement and risk adjustment in three ways. Relative to Driessen et al. (2012), we generalize the inception-only pricing restriction and provide an asymptotic framework that clarifies large-sample inference under overlapping funds. Relative to Korteweg and Nagel (2016), we show that both approaches are built on closely related cash-flow pricing restrictions, but that different aggregation and weighting choices imply different sample objectives and estimators. More broadly, the paper contributes to the growing literature on asset-pricing inference with illiquid, non-traded, and intermittently valued cash-flow streams, where the key challenge is not only estimation, but also determining which economically meaningful objects available data can identify with stability.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 develops the generalized semiparametric estimator and the

corresponding asymptotic inference framework. Section 4 studies finite-sample behavior in simulation and discusses implementation choices. Section 5 presents the empirical application to buyout and venture-capital funds. Section 6 concludes.

Appendix A analyzes the target shift and finite-sample stabilization under $\mathcal{T}$-averaging. Appendix B compares our simulation and empirical results to their Driessen et al. (2012) counterparts. Online Appendix C provides mathematical proofs.

2 Related literature

For private equity funds, performance evaluation under the SDF framework remains an active research topic (Driessen et al., 2012; Korteweg and Nagel, 2016, 2024; Giommetti and Jørgensen, 2021; Gupta and Nieuwerburgh, 2021; Boyer et al., 2023; Korteweg et al., 2024) alongside an expanding list of applications across other private asset classes. Specifically, the Driessen et al. (2012) estimator is heavily utilized to price human capital contracts, venture capital, real estate and private equity funds (Kroencke et al., 2013; Kim et al., 2015; Farrelly and Stevenson, 2019; Lee et al., 2019; Zhao, 2020; Tausch and Pietz, 2024), while the Korteweg and Nagel (2016) approach has found broad application spanning private equity, infrastructure, real estate investment trusts, collateralized loan obligations, impact funds and private debt (Giommetti and Jørgensen, 2021; Andonov et al., 2021; Riddiough and Wiley, 2022; Boyer et al., 2023; Cordell et al., 2023; Erel et al., 2024; Jeffers et al., 2024; Korteweg and Nagel, 2024). Previously, SDF performance evaluation was already applied for public stock markets by, e.g., Farnsworth et al. (2002) and for hedge fund returns by Li et al. (2016).

Within the private equity literature, SDF-based approaches can be broadly classified into parametric and semiparametric methods. Parametric models, such as those proposed by Franzoni et al. (2012) and Ang et al. (2018), explicitly specify the return dynamics and the underlying data generating process of private equity investments. While these methods can uncover latent factors or deal-level dynamics, they typically rely on strong distributional assumptions (e.g., log-normality) that may not fully capture the irregular and skewed nature of PE cash flows.

Consequently, semiparametric approaches have gained prominence, starting with the seminal cross-sectional GMM framework of Driessen et al. (2012). They estimate a linear factor model that minimizes the squared pricing errors of PE vintage year portfolios, explicitly evaluating discounted cash flows at fund inception. In a nutshell, the Driessen et al. (2012) estimator determines the estimated alpha and beta parameters of a linear factor model by minimizing the squared net present value (NPV) of all n fund cash flows in a least-squares optimization problem:

$$\left(\hat{\alpha}, \hat{\beta}\right) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} \left[NPV_i(\alpha, \beta)\right]^2 \tag{1}$$

with NPV for the ith fund

$$NPV_i(\alpha, \beta) = \sum_{t \in \{t_{0,i}, t_{0,i}+1, \dots\}} \frac{\text{CashFlow}_{i,t}}{\prod_{s=t_{0,i}+1}^{t} \left(1 + r_{f,s} + \alpha + \beta r_{\text{market},s}\right)}$$

with fund inception date $t_{0,i}$, risk-free rate $r_{f,s}$ and excess market return r_{market}. Here, the Driessen et al. (2012) approach relies on the identifying assumption that the model is correctly specified, so that $\mathbb{E}[NPV_i(\theta_0)] = 0$. Building on this, Korteweg and Nagel (2016) introduced the Generalized Public Market Equivalent (GPME) framework, shifting the estimation of the SDF to public market data while evaluating PE cash flows in a second step. Importantly, Korteweg and Nagel (2016) were the first to apply a spatial heteroskedasticity and autocorrelation consistent (SHAC) covariance matrix estimator (Conley, 1999) to account for the economic distance (cash flow overlap) between PE funds, moving beyond simple cross-sectional bootstrapping.

Further complementing this cash flow dimension, a strand of literature models the informational frictions and smoothing inherent in reported Net Asset Values (NAVs) (Brown et al., 2019; Jenkinson et al., 2020; Brown et al., 2023). Additionally, Hüther et al. (2022) leverage a similar generalized framework to evaluate private firms using credit market equivalents. Finally, for comprehensive methodological and empirical overviews of risk and return assessment in private equity, we refer to Korteweg (2019, 2023), who also discuss the popular Dimson (1979) time-series regression that employs lagged returns to correct for stale valuations.

Our paper's LMD framework contributes to this semiparametric literature by providing a loss-function formulation that encompasses the Driessen et al. (2012) approach and contextualizes the Korteweg and Nagel (2016) time-series analogue while supplying the rigorous increasing-domain asymptotic inference missing in earlier work (drawing on the spatial theory of Pötscher and Prucha (1997), Kim and Sun (2011), and Jenish and Prucha (2012)). While Gredil et al. (2020) highlighted that NPV-based measures are biased relative to per-period returns, we explicitly formalize this discrepancy as a "timing risk premium", a concept related to recent literature on discount-rate risk in secondary markets (Boyer et al., 2023).

3 Methodology

We now formalize a semiparametric framework for SDF estimation with pooled cash flows. Section 3.1 establishes the theoretical asset-pricing conditions and highlights how the choice of discounting date induces a "timing risk premium". Section 3.2 uses this insight to introduce a generalized least-mean-distance estimator that endogenizes the discounting dates. Section 3.3 builds an increasing-domain asymptotic framework to accommodate the spatial dependence of overlapping vintage years, and Section 3.4 describes large-sample inference. Section 3.5 demonstrates how this LMD estimator nests the framework of Driessen et al. (2012) as a special case and relates to Korteweg and Nagel (2016).

3.1 Asset pricing for pooled cash flows

Because private equity funds invest capital over multiple quarters, any fund-level pricing restriction must aggregate deal-level moments across different investment dates. This subsection asks under what conditions such a pooled cash-flow stream satisfies a valid stochastic-discount-factor pricing restriction. The central finding is

Symbol	Description
n	Number of cross-sectional units (funds or portfolios)
V	Number of distinct vintage years
n_v	Number of portfolios in vintage year v
v_i	Vintage year of fund/portfolio i
t_i^0	Inception date of fund i
J_i	Number of deals for fund i
$K_{i,j}$	Number of divestment cash flows for deal j of fund i
$cf_{i,j,t}$	Net cash flow of deal j in fund i at time t
$CF_{i,t}$	Net cash flow of fund i at time t
$P_{\tau,t,i}$	Time-τ price of a single time-t cash flow
$PV_i(\theta)$	Pooled present value
$LCV_{i,\tau}$	Latent continuation value of fund i at date τ
$T, T_i^{\max}$	Final period and maximum fund lifetime
τ	Generic discounting or revaluation date
$\mathcal{F}_\tau$	Information set (filtration) at time τ
$\mathrm{Inv}_{i,j}, \mathrm{Div}_{i,j,k}$	Investment and divestment cash flows
$t_{i,j}^{\mathrm{Inv}}, t_{i,j,k}^{\mathrm{Div}}$	Deal investment and divestment dates
$m_{i,j}^{\mathrm{NFV}}(\cdot)$	NFV-type theoretical moment
Ψ_t , M_t	Multi-period and single-period SDF
$\Psi_{\tau,t}(\theta)$	SDF ratio Ψ_t/Ψ_τ, parameterized by θ
$\theta \in \mathbb{R}^p$	SDF parameter vector (e.g., intercept and factor loadings)
θ_0	True parameter vector
$\delta_{i,j}$	Deal-level pricing error

Table 1: Key notation for asset pricing and pooled cash flows (Section 3.1).

that the choice of discounting date is not innocuous: discounting at or before the first investment date preserves a clean conditional zero-moment condition (Lemma 1, Part 1), whereas post-investment discounting or future-value compounding generally introduces a timing-related covariance wedge (Parts 2 and 3). The argument proceeds in three steps. First, we state a deal-level decomposition of fund cash flows (Assumption 1) and derive the timing result at the deal level (Lemma 1). Second, we aggregate to the fund level and show that pooled Net Present Value (NPV) moments satisfy a valid zero conditional moment restriction whenever discounting occurs at or before fund inception (Proposition 1). Third, we discuss the implications for Net Future Value (NFV) moments and the interpretation of alpha measured ex post. Throughout, the deal-level structure serves only as a latent accounting device used to justify the fund-level restriction; only pooled fund cash flows are required as observable inputs to the estimator.

For fund i, let $CF_{i,t}$ denote the observed net cash flow at time t, and define the fund inception date by $t_i^0 := \min_j t_{i,j}^{\mathrm{Inv}}$. All fund-level variables are constructed from the following deal-level data generation process.

Assumption 1. ***Deal-level decomposition.*** *For each fund i, there exist J_i underlying deals such that:*

1. *Deal j has exactly one negative investment cash flow* $\mathrm{Inv}_{i,j} < 0$ *at time* $t_{i,j}^{\mathrm{Inv}} \in \{0, 1, \ldots, T-1\}$.
2. *Deal j has a finite stream of positive divestment cash flows* $(\mathrm{Div}_{i,j,k})_{k=1,\ldots,K_{i,j}}$ *with* $\mathrm{Div}_{i,j,k} > 0$ *and* $t_{i,j,k}^{\mathrm{Div}} > t_{i,j}^{\mathrm{Inv}}$ *for all* k.
3. *Let $cf_{i,j,t}$ denote the deal-level cash flow of deal j in fund i at time t. Then the observed fund-level cash flow is the cross-deal aggregate*

$$CF_{i,t} = \sum_{j=1}^{J_i} cf_{i,j,t} \qquad \forall i, t.$$

4. *The fund has no cash flows before inception:* $CF_{i,t} = 0$ *for all* $t < t_i^0$.

Under our central asset-pricing restriction, each deal is correctly priced by the stochastic discount factor at its investment date. Formally, the deal-level pricing error satisfies

$$\mathbb{E}\left[\delta_{i,j} \,\middle|\, \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right] = 0 \quad \forall\, i, j \tag{2}$$

where the deal-level pricing error is defined by

$$\delta_{i,j} := \mathrm{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \frac{\Psi_{t_{i,j,k}^{\mathrm{Div}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}} \mathrm{Div}_{i,j,k} \tag{3}$$

and the multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ discounts a time-t cash flow to time 0 (and $\Psi_0 = 1$).

The next result characterizes how this deal-level pricing restriction depends on the choice of discounting or compounding date.

Lemma 1. *Suppose Equation* (2) *holds.*

1. *If* $\tau \le t_{i,j}^{\mathrm{Inv}}$, *then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{4}$$

2. *If* $\tau > t_{i,j}^{\mathrm{Inv}}$, *then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_\tau], \tag{5}$$

   which need not equal zero.

3. *Under the same pre-investment timing as Part (1), suppose additionally that the pricing error is compounded forward to a date* $t_{i,j}^{\mathrm{Future}} > t_{i,j}^{\mathrm{Inv}}$. *If* $\tau \le t_{i,j}^{\mathrm{Inv}} < t_{i,j}^{\mathrm{Future}}$, *then define the NFV-type moment*

$$m_{i,j}^{\mathrm{NFV}}(\tau, t_{i,j}^{\mathrm{Future}}) := \frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_\tau}\,\delta_{i,j}.$$

Its expected value

$$\mathbb{E}\left[m_{i,j}^{\mathrm{NFV}}(\tau, t_{i,j}^{\mathrm{Future}}) \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) \mid \mathcal{F}_\tau\right] \tag{6}$$

generally contains a conditional covariance term and therefore need not equal zero.

Proof. Consider part (1). Applying the law of iterated expectations,[1] since $\tau \leq t_{i,j}^{\mathrm{Inv}}$, the SDF ratio is known at $t_{i,j}^{\mathrm{Inv}}$ and can be factored out yielding

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right]}_{=0 \text{ by Equation (2)}} \mid \mathcal{F}_\tau\right] = 0.$$

For part (2), since $t_{i,j}^{\mathrm{Inv}} < \tau$, the ratio $\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau$ is $\mathcal{F}_\tau$-measurable, so

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_\tau].$$

Since some divestment cash flows of deal j may have been paid by date τ, $\delta_{i,j}$ is partially $\mathcal{F}_\tau$-measurable, and its conditional expectation given $\mathcal{F}_\tau$ generally reflects realized outcomes rather than the zero-mean restriction at $t_{i,j}^{\mathrm{Inv}}$.

For part (3), using the law of iterated expectations with an intermediate conditioning on $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$,

$$\begin{aligned}\mathbb{E}\left[m_{i,j}^{\mathrm{NFV}}(\tau, t_{i,j}^{\mathrm{Future}}) \mid \mathcal{F}_\tau\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right] \mid \mathcal{F}_\tau\right] \\ &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) \mid \mathcal{F}_\tau\right]\end{aligned}$$

where we used the conditional covariance identity and Equation (2). □

Part (3) of Lemma 1 shows that compounding deal cash flows to a future date generally destroys the zero conditional moment restriction because the resulting expression contains a covariance between the pricing error and future SDF realizations. We refer to this non-zero expected value as the *timing wedge*. Economically, it is the time-τ price of an investment-date covariance term between continuation SDF realizations and the discounted divestment cash flows over the illiquid holding period. We label this component the **Timing Risk Premium (TRP)** because it captures compensation for systematic timing exposure, i.e., exposure to the states in which realization occurs and discount rates are realized.

[1] At the boundary $\tau = t_{i,j}^{\mathrm{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau = 1$ is a trivial constant, so the covariance with $\delta_{i,j}$ is zero and the same argument applies.

To facilitate the economic interpretation of the TRP, let us briefly analyze Equation 6 with only one divestment cash flow per deal ($K_{i,j} = 1$). In this case, the covariance term reads

$$\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}},\ \mathrm{Inv}_{i,j} + \frac{\Psi_{t_{i,j,k}^{\mathrm{Div}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}\mathrm{Div}_{i,j,k} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) = \mathrm{Cov}\left(\prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j}^{\mathrm{Future}}} M_u,\ \mathrm{Div}_{i,j,k}\prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j,k}^{\mathrm{Div}}} M_u \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right)$$

Thus, for $K_{i,j} = 1$, the covariance is naturally expected to be largest when $t_{i,j}^{\mathrm{Future}}$ is equal to $t_{i,j,k}^{\mathrm{Div}}$, since the continuation SDF and the discounted divestment payoff then load on the same SDF path. This ranking is not automatic, however, and requires additional restrictions on the post-divestment SDF dynamics (e.g, $\Psi_{t_{i,j}^{\mathrm{Future}}}/\Psi_{t_{i,j}^{\mathrm{Div}}}$ being conditionally mean-independent of $\mathcal{F}_{t_{i,j}^{\mathrm{Div}}}$ given $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$).

To conclude our analysis of Part 3 of Lemma 1, NFV-type moments are inappropriate as primitive ex-ante pricing moments unless the TRP term is modeled explicitly. They can still be informative for ex-post benchmarking, but the resulting performance measure must then be interpreted as a composite object that mixes true abnormal performance (skill) with a systematic timing exposure-related compensation (TRP).

Proposition 1. ***Price of a pooled cash-flow stream at or before fund inception.*** *If $\tau \leq t_i^0$, then*

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,\middle|\,\mathcal{F}_\tau\right] = 0. \tag{7}$$

Proof. If $\tau \leq t_i^0$, then $\tau \leq t_{i,j}^{\mathrm{Inv}}$ for every deal j in fund i. Using the deal-level decomposition of the pooled cash-flow stream and Equation (3),

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,\middle|\,\mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J_i}\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right]. \tag{8}$$

Applying Equation (4) term-by-term yields

$$\mathbb{E}\left[\sum_{j=1}^{J_i}\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right] = 0. \tag{9}$$

□

Proposition 1 provides the clean benchmark for identification: pooled NPV moments are theoretically well founded when discounting occurs at or before fund inception. Once $\tau > t_i^0$, some underlying deals are active or already partially realized, so the corresponding pooled pricing error incorporates the timing wedge. The choice of discounting dates is therefore not innocuous. Section 3.2 formalizes this choice by defining an estimator that explicitly specifies which discounting dates are used. When it restricts to dates $\tau \leq t_i^0$, the pooled fund-level pricing error inherits

a valid zero conditional moment. When it includes dates $\tau > t_i^0$, it generally targets a tilted criterion that embeds the Timing Risk Premium characterized by Equation (6).

Critically, this Timing Risk Premium is not solely relevant for ex-post benchmarking. It also fundamentally drives the large-sample and small-sample properties of our generalized LMD estimator. As we demonstrate in later sections, averaging pricing errors over extended horizons deliberately injects this structural timing wedge to control severe finite-sample estimator bias and variance.

3.2 Least-Mean-Distance estimator

Symbol	Description
$\epsilon_{\tau,i}$	Fund pricing error at discounting date τ
$\bar{\epsilon}_i, \bar{\epsilon}_i^{(H)}$	Weighted average pricing error (generic and horizon-specific)
H	Horizon parameter
$\mathcal{T}_i, \mathcal{T}_{i,H}$	Set of discounting dates (generic and horizon-specific)
w_i	Weighting factor for fund i
$L(\cdot)$	Loss function (e.g., $L(x) = x^2$)
$Q_H(\theta)$	Horizon-specific population objective function
$Q_n(\theta), Q_{n,H}(\theta)$	Sample objective function (generic and horizon-specific)
$s_i^{(H)}(\theta)$	Unit score contribution
$d_{i,j}$	Economic distance between units i and j
$K(\cdot), k_{i,j}$	Symmetric SHAC kernel function and generic weights
b_V	SHAC bandwidth parameter
$\hat{\theta}, \hat{\theta}_H$	LMD Estimator (generic and horizon-specific targets)
$\theta_H^\dagger$	Horizon-specific pseudo-true parameter vector
$\mathcal{H}_H$	Hessian of population objective at $\theta_H^\dagger$
Λ_H	Long-run covariance matrix of the score
Σ_H	Asymptotic covariance matrix of $\hat{\theta}_H$
W_H	Wald statistic for linear hypotheses
R, r	Linear constraint matrix and vector

Table 2: Key notation for the LMD estimator and asymptotic inference (Sections 3.2–3.4).

In this subsection, we define a generalized fund-level criterion that averages pricing errors across a user-specified set of discounting dates $\mathcal{T}_i$. This nests the inception-only NPV criterion of Driessen et al. (2012) as a special case and allows us to study how horizon averaging trades structural validity against finite-sample regularization. Throughout, only pooled fund cash flows are observed, whereas the deal-level decomposition from Section 3.1 remains a latent theoretical device.

The stochastic discount factor $\Psi_{\tau,t} = \Psi_t/\Psi_\tau$ is used to calculate the time-τ price

$P_{\tau,t,i}$ of a **single** time-t cash flow of any given PE fund i

$$P_{\tau,t,i} := \Psi_{\tau,t} \cdot CF_{i,t} = \frac{\Psi_t}{\Psi_\tau} \cdot CF_{i,t} \qquad \forall \quad \tau, t, i \tag{10}$$

with again multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ and $\Psi_0 = 1$. As SDFs are commonly parameterized by a vector $\theta \in \mathbb{R}^p$, i.e., $\Psi_{\tau,t} \equiv \Psi_{\tau,t}(\theta)$, our goal is to find an estimation method for the true parameter vector θ_0.

For each fund i and all points τ within a common fund lifetime, the empirical pricing error $\epsilon_{\tau,i}(\theta)$ of **all** realized-to-date fund cash flows is calculated as a time-τ revaluation of the entire cash-flow stream

$$\epsilon_{\tau,i}(\theta) := \sum_{t=0}^{T} P_{\tau,t,i}(\theta) \qquad \forall \quad \tau, i \tag{11}$$

Note that for $t < \tau$, the terms $\frac{\Psi_t}{\Psi_\tau} CF_{i,t}$ are compounded past cash flows, not discounted future cash flows, so the term "present value" is no longer accurate for $\tau > t_i^0$. For $\tau > t_i^0$, $\epsilon_{\tau,i}(\theta)$ is not a continuation value but a time-τ revaluation of the entire realized cash-flow stream, and therefore generally inherits the timing wedge characterized in Section 3.1. This time-τ revaluation $\epsilon_{\tau,i}(\theta)$ must be sharply distinguished from the latent continuation value $LCV_{i,\tau}(\theta) := \mathbb{E}[\sum_{t=\tau}^{T} (\Psi_t/\Psi_\tau) CF_{i,t} \mid \mathcal{F}_\tau]$, which conditions on information at τ and is not directly observed.

To analyze the effect of different discounting dates τ on the bias and variance of the estimated SDF parameters, we define the generalized criterion $\bar{\epsilon}_i(\theta)$ that averages pricing errors over the set $\mathcal{T}_i$:

$$\bar{\epsilon}_i(\theta) = w_i \cdot \frac{1}{\mathrm{card}(\mathcal{T}_i)} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i}(\theta) \qquad \forall \quad i \tag{12}$$

where w_i normalizes fund-level pricing errors before the loss is applied; it is not an observation weight in the outer sample average. Depending on the application, w_i may normalize by invested capital, by aggregate vintage-year capital, by a macroeconomic deflator, or be set to one when cash flows are already expressed in comparable scale units.

The role of $\mathcal{T}_i$ The set $\mathcal{T}_i$ controls which discounting dates enter the averaged pricing error $\bar{\epsilon}_i(\theta)$ in Equation (12) (see Figure 1). Averaging over multiple dates in $\mathcal{T}_i$ was originally motivated by the "exploding alpha" problem first described by Driessen et al. (2012): when the initial cash flow is very small, an arbitrarily large alpha can discount all subsequent cash flows close to zero, rendering the beta factors irrelevant and distorting estimation. Averaging over $\mathcal{T}_i$ mitigates this degeneracy. The relevant horizon family is parameterized as

$$\mathcal{T}_{i,H} = \left\{ t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max}) \right\}, \qquad H \geq 0. \tag{13}$$

This formulation encompasses two economically distinct cases:

- $H = 0$: If $\mathcal{T}_i = \{t_i^0\}$, the estimator uses an inception-only NPV criterion, is anchored in the pooled pricing restriction from Section 3.1, and targets the structural parameter θ_0.

- $H > 0$: If $\mathcal{T}_i = \mathcal{T}_{i,H}$ with $H > 0$, the estimator is a generalized horizon-averaged criterion and targets the pseudo-true parameter $\theta_H^\dagger$. It provides finite-sample regularization benefits but incorporates a potential timing distortion.

Larger values of the horizon H offer more regularization, but a larger departure from the structural benchmark with zero expected value.

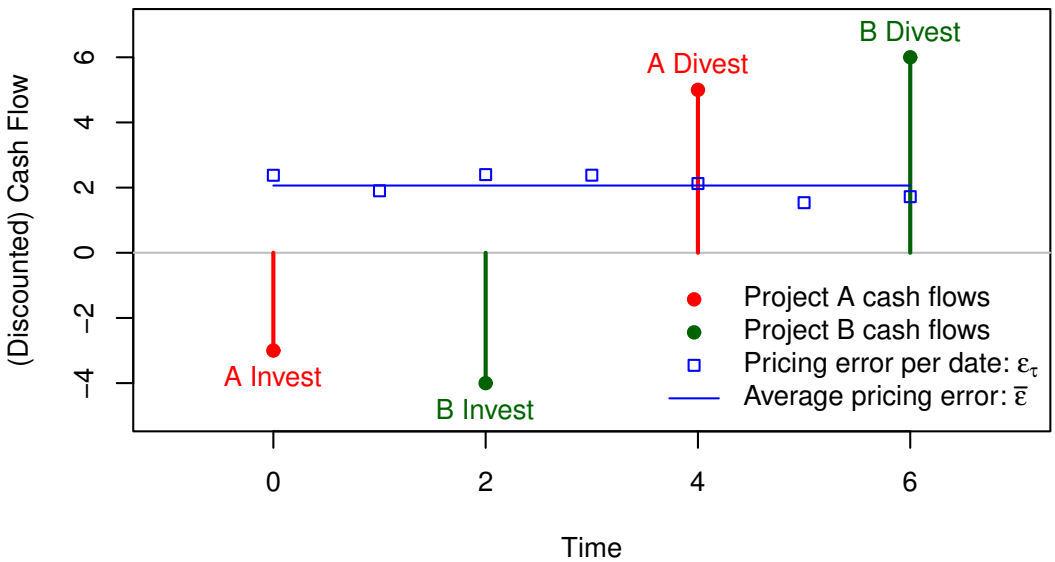

Figure 1: How to calculate and interpret the average pricing error introduced by Equation (12): The time index t is relevant for the net cash flows (black dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation (12) (solid blue line). In this example, $\text{card}(\mathcal{T}_i) = 7$, i.e., the number of blue boxes.

Remark 1 (Target parameter under horizon averaging)**.** *For $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, the estimator targets the horizon-specific pseudo-true parameter $\theta_H^\dagger = \arg\max_\theta Q_H(\theta)$. This coincides with the structural parameter θ_0 only when $H = 0$. For $H > 0$, the estimator trades structural fidelity against finite-sample regularization by averaging pricing errors over discounting dates that may incorporate the timing wedge from Section 3.1.*

Restriction used for identification and asymptotic theory. In the consistency analysis below, we specialize the generalized criterion to $\mathcal{T}_i = \{t_i^0\}$. Under Proposition 1, this yields a valid pooled fund-level zero-moment restriction at the structural parameter θ_0.

Remark 2. *This restriction guarantees that $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$ under the true parameter. For the quadratic loss $L(x) = x^2$, the population objective $Q_0(\theta) = -\mathbb{E}[\bar{\epsilon}_i(\theta)^2] = -\text{Var}[\bar{\epsilon}_i(\theta)] - (\mathbb{E}[\bar{\epsilon}_i(\theta)])^2$ involves both variance and bias terms. Consistency then requires that θ_0 uniquely minimizes this composite criterion (see Proposition 3).*

Because our estimator is defined through the optimization of a sample criterion, it is an extremum estimator in the sense of Newey and McFadden (1994, Eq. 1.1), with $\hat{\theta} = \arg\max_{\theta\in\Theta} Q_n(\theta)$.

Concretely, our Least Mean Distance (LMD) estimator (Pötscher and Prucha, 1997, Equation 7.1) minimizes the average loss of $\bar{\epsilon}_i(\theta)$. For consistency with the extremum-estimator convention, we will technically maximize the average negative loss of $\bar{\epsilon}_i(\theta)$, which is equivalent.

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \ \frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big) \equiv \arg\max_{\theta\in\Theta} \ Q_n(\theta) \tag{14}$$

with

$$Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big)$$

where $L : \mathbb{R}^1 \to \mathbb{R}^1$ denotes a loss function, e.g., $L(x) = (x-0)^2$ in the case of nonlinear least squares. Throughout the paper, the weighted average fund pricing error $\bar{\epsilon}_i \equiv \bar{\epsilon}_i(\theta)$ is regarded as a nonlinear random function of the SDF parameter θ.

3.3 Asymptotic framework

The large-sample analysis is stated for the horizon-indexed estimator

$$\hat{\theta}_H = \arg\max_{\theta\in\Theta} Q_{n,H}(\theta), \qquad Q_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n} L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \tag{15}$$

where

$$\bar{\epsilon}_i^{(H)}(\theta) := w_i \cdot \frac{1}{\mathrm{card}(\mathcal{T}_{i,H})} \sum_{\tau\in\mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta)$$

and

$$\mathcal{T}_{i,H} = \left\{t_i^0, t_i^0+1, \ldots, \min(t_i^0+12H, T_i^{\max})\right\}.$$

Throughout this subsection, the horizon parameter $H \geq 0$ is fixed while $n \to \infty$.

The notation in Equation (15) makes explicit that the estimator and its probability limit generally depend on the chosen horizon H. This distinction is essential in the present setting because the pooled inception-based pricing restriction yields a structural zero-moment condition only for $H = 0$, whereas horizon averaging with $H > 0$ may introduce the timing wedge discussed in Section 3.1.

Population criterion and target parameter. Define the horizon-specific population objective by

$$Q_H(\theta) := \mathbb{E}[Q_{n,H}(\theta)] = -\mathbb{E}\Big[L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big)\Big]. \tag{16}$$

Under the ergodicity implied by the increasing-domain framework (Assumption 3), the Cesàro average $n^{-1}\sum_{i=1}^{n}\mathbb{E}[L(\bar{\epsilon}_i^{(H)}(\theta))]$ converges to the single-unit expectation as $V \to \infty$. The corresponding population target is the unique maximizer

$$\theta_H^{\dagger} := \arg\max_{\theta\in\Theta} Q_H(\theta). \tag{17}$$

Remark 3 (Structural versus horizon-specific target)**.** *The asymptotic theory below is intentionally formulated around $\theta_H^\dagger$, not around θ_0. This distinction is substantive rather than cosmetic. For $H = 0$, the pooled inception-only criterion coincides with the structural pricing restriction of Section 3.1, and hence $\theta_0^\dagger = \theta_0$. For $H > 0$, the criterion generally targets a pseudo-true parameter $\theta_H^\dagger \neq \theta_0$, because averaging over $\tau > t_i^0$ may embed the timing wedge at the population level.*

This formulation keeps the asymptotic statements aligned with the estimator actually implemented in the empirical analysis. It also cleanly separates two conceptually distinct questions: (i) the large-sample behavior of the estimator relative to its own criterion, and (ii) the economic interpretation of that criterion relative to the structural parameter θ_0.

3.3.1 Vintage-year increasing-domain asymptotics

By the value-additivity principle (Hansen and Richard, 1987), the SDF restriction can be formulated for pooled cash flows at different aggregation levels. In the empirical implementation, we form vintage-year portfolios as cross-sectional units, because this reduces idiosyncratic fund-level noise while preserving economically meaningful variation across macro environments.

Assumption 2. *For each vintage year v, we pool fund cash flows to form n_v portfolios, so that*

$$n = \sum_{v=1}^{V} n_v.$$

The boundary cases are: (i) single-fund portfolios and (ii) one portfolio per vintage year. Both equal-weighted and value-weighted portfolio constructions are admissible.

Without loss of generality, we continue to refer to the resulting cross-sectional units as funds. Let $v_i := \mathrm{YEAR}(t_i^0)$ denote the vintage year of unit i.

The pricing error $\bar{\epsilon}_i^{(H)}(\theta)$ is indexed by both unit identity and vintage year. Accordingly, the data are naturally represented as an increasing-domain random field rather than as a standard iid cross-section or a conventional long time series. Economically, dependence arises because adjacent vintage years are exposed to overlapping macroeconomic conditions and partially overlapping realized cash-flow windows.

Assumption 3 (Vintage-year increasing-domain asymptotics)**.** *As $n \to \infty$:*

1. *The number of distinct vintage years satisfies $V \to \infty$.*
2. *The number of units per vintage year is uniformly bounded: there exists $\bar{n} < \infty$ such that $n_v \leq \bar{n}$ for all v.*
3. *Fund lifetimes are uniformly bounded: there exists $\bar{T} < \infty$ such that $T_i^{\max} - t_i^0 \leq \bar{T}$ for all i.*
4. *The economic distance between units i and j is measured by*

$$d_{i,j} = |v_i - v_j| + \rho_0 \mathbf{1}_{\{i \neq j\}}, \qquad \rho_0 > 0.$$

Assumption 3 imposes increasing-domain asymptotics and rules out infill asymptotics in the sense of Jenish and Prucha (2012). The role of $\rho_0 > 0$ is to prevent the field from becoming asymptotically dense at a fixed location. This is economically natural here: asymptotic information is generated by observing additional vintage years, not by observing an ever denser sample from a fixed macroeconomic environment.

A useful implication of Assumption 3(2) is that the total number of observations n and the number of vintages V are of the same order. Indeed,

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n}V,$$

so $n \asymp V$. Consequently, $\sqrt{n}$- and $\sqrt{V}$-rates are asymptotically equivalent up to a constant factor. Put differently, statistical information grows with the span of vintage years rather than with within-vintage clustering.

This increasing-domain perspective differs from the infill framework of Driessen et al. (2012), who fix the number of portfolios and let the number of underlying funds per portfolio diverge. Their asymptotic device is appropriate for consistency conditional on a fixed macroeconomic environment. By contrast, the present framework is designed for unconditional identification and inference when new vintages generate new realizations of macroeconomic conditions.

3.3.2 Uniform convergence of the criterion

The first step is a uniform law of large numbers for the horizon-specific sample criterion $Q_{n,H}(\theta)$.

Assumption 4 (Parameter space)**.** *The parameter space $\Theta \subset \mathbb{R}^p$ is compact.*

Assumption 5 (Primitive regularity conditions for uniform convergence)**.** *For each fixed H:*

1. *The array*
$$\left\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1, \ldots, n\right\}$$
is near-epoch dependent (NED) on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$, uniformly in $\theta \in \Theta$ (that is, the NED approximation constants are bounded uniformly over $\theta \in \Theta$ and i).

2. *The loss function $L(\cdot)$ is continuous and Lipschitz on the relevant support.*

3. *There exists $p > 2$ such that*
$$\sup_{\theta\in\Theta} \mathbb{E}\left[\left|L\left(\bar{\epsilon}_i^{(H)}(\theta)\right)\right|^p\right] < \infty$$
uniformly in i.

4. *The criterion is stochastically equicontinuous in θ.*

Assumption 5 is stated at a deliberately primitive level: dependence is imposed on the underlying horizon-specific pricing errors, and the loss inherits the required regularity through continuity and Lipschitz continuity. This is preferable to assuming a law of large numbers directly for the transformed criterion because it ties the statistical assumptions more closely to the economic structure of the data.

Proposition 2 (Uniform law of large numbers)**.** *Under Assumptions 2, 3, and 5,*

$$\sup_{\theta \in \Theta} |Q_{n,H}(\theta) - Q_H(\theta)| \xrightarrow{p} 0.$$

Outline of Proof. The result follows from a uniform law of large numbers for NED random fields on α-mixing bases; see, for example, Jenish and Prucha (2009, Theorem 6). The bounded-lifetime and bounded-within-vintage assumptions ensure that cross-sectional overlap remains local in vintage-year distance, while the moment and equicontinuity conditions deliver uniform stochastic control over $\theta \in \Theta$. □

Economically, Proposition 2 says that when the sample spans sufficiently many distinct vintages, the empirical criterion stabilizes around its population counterpart. This remains true whether the horizon is structural ($H = 0$) or regularized ($H > 0$); what changes across horizons is the population target itself.

3.3.3 Consistency

Assumption 6 (Identification of the horizon-specific target)**.** *For each fixed H, the population criterion $Q_H(\theta)$ has a unique maximizer*

$$\theta_H^\dagger = \arg\max_{\theta \in \Theta} Q_H(\theta).$$

Proposition 3 (Consistency)**.** *Under Assumptions 2, 3, 4, 5, and 6,*

$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger \qquad \text{as } n \to \infty.$$

Outline of Proof. The claim follows from the standard extremum-estimator argument of Newey and McFadden (1994, Theorem 2.1). Proposition 2 yields uniform convergence of $Q_{n,H}$ to Q_H, Assumption 6 provides a unique maximizer, and compactness of Θ (Assumption 4) together with continuity of the discount-factor mapping $\Psi_{\tau,t}(\theta)$ and the loss function deliver the remaining conditions. □

Corollary 1 (Structural consistency under inception-only pricing)**.** *Suppose, in addition, that $H = 0$ and that the pooled inception-only pricing restriction of Proposition 1 holds at the structural parameter θ_0. Then*

$$\theta_0^\dagger = \theta_0 \qquad \text{and hence} \qquad \hat{\theta}_0 \xrightarrow{p} \theta_0.$$

Corollary 1 isolates the special case in which the estimator converges to the structural SDF parameter. This separation is useful empirically because it prevents the reader from conflating consistency of the extremum estimator with structural validity of a horizon-averaged criterion.

3.3.4 Central limit theorem

Building on the consistency result in Proposition 3, we now establish the asymptotic distribution of the estimator. Like consistency, the central limit theorem is formulated around the horizon-specific target $\theta_H^\dagger$. This requires a standard local-convergence condition on the sample Hessian.

Assumption 7 (Local Hessian convergence)**.** *For each fixed H, there exists an open neighborhood $\mathcal{N}_H$ of $\theta_H^\dagger$ such that*

$$\sup_{\theta \in \mathcal{N}_H} \|\nabla_{\theta\theta} Q_{n,H}(\theta) - \nabla_{\theta\theta} Q_H(\theta)\| \xrightarrow{p} 0.$$

Assumption 7 is the standard sample-Hessian convergence condition used in extremum-estimator central limit theorems (e.g., Newey and McFadden, 1994). In our increasing-domain framework, it follows from the same random-field LLN argument as Proposition 2, provided that the Hessian contributions inherit the required NED, moment, and stochastic-equicontinuity properties from the pricing errors $\bar{\epsilon}_i^{(H)}(\theta)$ and the loss function $L(\cdot)$.

Proposition 4 (Asymptotic normality around the horizon-specific target)**.** *Under the conditions of Proposition 3 and Assumption 7, suppose additionally that:*

1. *$\theta_H^\dagger$ lies in the interior of Θ;*
2. *$Q_H(\theta)$ is twice continuously differentiable in a neighborhood of $\theta_H^\dagger$;*
3. *the Hessian*
$$\mathcal{H}_H := \nabla_{\theta\theta} Q_H(\theta_H^\dagger)$$
is nonsingular; and
4. *a central limit theorem holds for the score contributions*
$$s_i^{(H)}(\theta) := \nabla_\theta L\left(\bar{\epsilon}_i^{(H)}(\theta)\right).$$

Then

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_H),$$

where

$$\Sigma_H = \mathcal{H}_H^{-1} \Lambda_H \mathcal{H}_H^{-1}, \qquad \Lambda_H := \lim_{n\to\infty} n \cdot \mathrm{Var}\left[\nabla_\theta Q_{n,H}(\theta_H^\dagger)\right].$$

Outline of Proof. The result follows from the extremum-estimator CLT of Newey and McFadden (1994, Theorem 3.1) combined with a central limit theorem for NED random fields under increasing-domain asymptotics; see Jenish and Prucha (2012, Theorem 4). The nonstandard feature in the present application is the vintage-year dependence of the score contributions. Assumptions 3 and 5 deliver the weak dependence and moment conditions needed to apply the random-field CLT (where compactness of Θ is inherited from Assumption 4). □

Corollary 2 (Structural asymptotic normality in the inception-only case)**.** *Under the additional conditions of Corollary 1,*

$$\sqrt{n}\left(\hat{\theta}_0 - \theta_0\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_0).$$

Remark 4 (Interpretation of the asymptotic center)**.** *For $H > 0$, Proposition 4 is centered at $\theta_H^\dagger$, not at θ_0. Therefore, large-sample inference in the horizon-averaged case pertains to the pseudo-true parameter defined by the criterion itself. Structural interpretation relative to θ_0 requires the target-shift analysis developed later in the paper.*

3.4 Large sample inference

Building on the asymptotic normality established above, we now turn to the practical implementation of large-sample inference. Proposition 4 implies the familiar sandwich form

$$\Sigma_H = \mathcal{H}_H^{-1} \Lambda_H \mathcal{H}_H^{-1}.$$

We now describe consistent estimators of the two building blocks $\mathcal{H}_H$ and Λ_H.

Score contributions. It is useful to define the score contribution of unit i explicitly as

$$s_i^{(H)}(\theta) := \nabla_\theta L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = L'\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) \cdot \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta). \tag{18}$$

For the quadratic loss $L(x) = x^2$, Equation (18) simplifies to

$$s_i^{(H)}(\theta) = 2\bar{\epsilon}_i^{(H)}(\theta) \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta).$$

This explicit definition makes clear that the long-run covariance matrix Λ_H is the covariance of the criterion score, not the covariance of the pricing errors themselves.

Hessian estimator. A sample analogue of the Hessian is

$$\hat{\mathcal{H}}_H := \nabla_{\theta\theta} Q_{n,H}(\hat{\theta}_H) = -\frac{1}{n} \sum_{i=1}^{n} \nabla_{\theta\theta} L\Big(\bar{\epsilon}_i^{(H)}(\hat{\theta}_H)\Big). \tag{19}$$

When analytic derivatives are unavailable, the derivatives entering Equation (19) can be computed numerically. In practice, central finite differences are typically adequate, although automatic differentiation or quasi-Newton approximations may also be used. For a methodological paper, the exact numerical routine is secondary; what matters asymptotically is that $\hat{\mathcal{H}}_H$ consistently estimates $\mathcal{H}_H$. In our empirical applications, we use the central difference approximations of (Eu, 2017, Algorithm 2).

SHAC estimator of the long-run covariance matrix. Because the score contributions exhibit vintage-year dependence, a heteroskedasticity- and autocorrelation-consistent covariance estimator is required. We use the spatial HAC estimator of (Kim and Sun, 2011, Equation 2) adapted to the vintage-year metric:

$$\hat{\Lambda}_H = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} k_{i,j}\, s_i^{(H)}(\hat{\theta}_H) s_j^{(H)}(\hat{\theta}_H)^\top, \qquad k_{i,j} = K\left(\frac{d_{i,j}}{b_V}\right), \tag{20}$$

where $d_{i,j}$ is the vintage-year distance from Assumption 3 and $K(\cdot)$ is a symmetric kernel satisfying the standard SHAC regularity conditions.

A convenient choice is the Bartlett kernel

$$K_{BT}(x) = \max(0, 1 - |x|).$$

The associated bandwidth sequence b_V should satisfy

$$b_V \to \infty, \qquad \frac{b_V}{V} \to 0, \tag{21}$$

which is the standard condition ensuring consistent long-run covariance estimation under weak dependence. Equation (21) is a population-theoretic requirement. In empirical work, one may implement a finite truncation rule for b_V and assess robustness across reasonable bandwidth choices.

This distinction between theory and implementation is important. A fixed bandwidth may be useful in a finite sample and can be justified as an approximation when dependence becomes negligible beyond a small vintage-year lag. However, under generic weak dependence, consistency of SHAC is established under a diverging bandwidth sequence such as Equation (21).

Asymptotic covariance matrix, standard errors, and Wald tests. The estimated asymptotic covariance matrix is

$$\widehat{\mathrm{Avar}}(\hat{\theta}_H) = \frac{1}{n}\hat{\mathcal{H}}_H^{-1}\hat{\Lambda}_H\hat{\mathcal{H}}_H^{-1}. \tag{22}$$

Accordingly, the vector of asymptotic standard errors is

$$\mathrm{SE}(\hat{\theta}_H) = \sqrt{\mathrm{diag}\left[\widehat{\mathrm{Avar}}(\hat{\theta}_H)\right]}. \tag{23}$$

This presentation is algebraically equivalent to the usual sandwich formula, but it is more transparent because the $1/n$ scaling is stated directly at the covariance-matrix level.

For linear hypotheses of the form

$$\text{Hypothesis}_0 : \mathcal{R}\theta = r, \qquad \text{Hypothesis}_1 : \mathcal{R}\theta \neq r,$$

the Wald statistic is

$$W_H = (\mathcal{R}\hat{\theta}_H - r)^\top \left[\mathcal{R}\widehat{\mathrm{Avar}}(\hat{\theta}_H)\mathcal{R}^\top\right]^{-1} (\mathcal{R}\hat{\theta}_H - r), \tag{24}$$

which is asymptotically χ^2_q under Hypothesis_0, where q is the number of linear restrictions.

Remark 5 (Interpretation of inference when $H > 0$)**.** *When $H > 0$, the standard errors and Wald tests in Equations* (23) *and* (24) *are asymptotically valid for the pseudo-true parameter $\theta_H^\dagger$. They should therefore be interpreted as inference on the horizon-specific criterion minimizer, not automatically as structural tests about θ_0.*

3.5 Comparison to similar estimators

Our Least-Mean-Distance (LMD) estimator belongs to the broad class of semiparametric nonlinear extremum estimators in the sense of Pötscher and Prucha (1997). Its main advantage in the present setting is that it accommodates pooled fund-level pricing errors, flexible horizon averaging through $\mathcal{T}_{i,H}$, and increasing-domain asymptotics indexed by vintage years, all within a comparatively transparent objective function.

The relationship to the existing private-equity asset-pricing literature is best understood at the level of objective functions and aggregation schemes. The estimator is most closely connected to Driessen et al. (2012), while it is more loosely related to the time-series-like GMM construction of Korteweg and Nagel (2016). The relevant point is not that all three estimators are identical, but rather that they occupy nearby positions in the same semiparametric estimation landscape.

3.5.1 Connection to Driessen et al. (2012)

The connection to Driessen et al. (2012) is especially close. Under the inception-only restriction, equal normalization, and quadratic loss, the LMD criterion coincides with the identity-weighted one-step cross-sectional GMM objective used in their framework. To bridge the gap, let us rewrite the NPV_i from Equation 1 in our more general SDF notation as $PV_i(\theta) = \sum_{t \in \{t_{0,i}, t_{0,i}+1, \dots\}} \Psi_{t_{0,i},t}(\theta) CF_t$.

Proposition 5 (Connection to the Driessen et al. (2012) objective)**.** *Suppose that:*

1. $\mathcal{T}_i = \{t_i^0\}$ *for all* i*,*
2. $w_i = 1$ *for all* i*, and*
3. $L(x) = x^2$*.*

Then the LMD estimator solves

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \left(PV_i(\theta)\right)^2,$$

which is equivalent to the identity-weighted cross-sectional GMM objective in Driessen et al. (2012).

Proof. Under $\mathcal{T}_i = \{t_i^0\}$, the averaged pricing error reduces to the inception-based pooled net present value

$$\bar{\epsilon}_i(\theta) = \epsilon_{t_i^0,i}(\theta) = \sum_{t=0}^{T} \frac{\Psi_t(\theta)}{\Psi_{t_i^0}(\theta)} CF_{i,t} = PV_i(\theta).$$

With $w_i = 1$ and $L(x) = x^2$, Equation (14) becomes

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \left(PV_i(\theta)\right)^2.$$

Writing the vector of pooled pricing errors as $e(\theta) = (PV_1(\theta), \dots, PV_n(\theta))^\top$, the objective is proportional to $e(\theta)^\top e(\theta)$, which is the identity-weighted one-step GMM criterion. □

3.5.2 Connection to Korteweg and Nagel (2016)

The relation to Korteweg and Nagel (2016) is closely related at the level of discounted cash-flow moments, but it is not a literal nesting result. Their estimator applies a GMM quadratic form to an aggregated mean moment vector of discounted private-equity and benchmark cash flows, whereas our LMD criterion applies the loss at the unit level before averaging. In their empirical implementation, the weighting matrix places zero weight on the private-equity pricing-error component, so the SDF parameters are identified by benchmark moments alone and the private-equity pricing error is evaluated after estimation. Accordingly, Korteweg and Nagel (2016) is best viewed as a closely related aggregated-moment GMM analogue rather than as a special case of Equation (14).

The difference can be formalized in a scalar bridge case. Let $m_i(\theta)$ denote a unit-level pricing error constructed from the same primitive discounted cash flows, and define

$$Q_N^{\mathrm{LMD}}(\theta) := \frac{1}{N}\sum_{i=1}^{N} m_i(\theta)^2, \qquad Q_N^{\mathrm{Agg}}(\theta) := \big(\bar{m}_N(\theta)\big)^2, \qquad \bar{m}_N(\theta) := \frac{1}{N}\sum_{i=1}^{N} m_i(\theta).$$

This scalar bridge case is used only to isolate the aggregation-order distinction in its simplest form. A vector analogue is immediate, but in the full Korteweg and Nagel (2016) setup it would additionally conflate aggregation order with moment selection and the weighting-matrix choice that excludes the private-equity pricing moment from parameter estimation.

Proposition 6 (Loss before aggregation versus aggregation before loss)**.** *For every θ,*

$$Q_N^{\mathrm{LMD}}(\theta) = Q_N^{\mathrm{Agg}}(\theta) + \frac{1}{N}\sum_{i=1}^{N}\big(m_i(\theta) - \bar{m}_N(\theta)\big)^2.$$

Hence $Q_N^{\mathrm{LMD}}(\theta) \geq Q_N^{\mathrm{Agg}}(\theta)$, with equality if and only if $m_i(\theta) = \bar{m}_N(\theta)$ for all i.

Proof. This is the standard variance decomposition:

$$\frac{1}{N}\sum_{i=1}^{N} m_i(\theta)^2 = \big(\bar{m}_N(\theta)\big)^2 + \frac{1}{N}\sum_{i=1}^{N}\big(m_i(\theta) - \bar{m}_N(\theta)\big)^2. \qquad \square$$

Proposition 6 shows that the two criteria coincide only under restrictive homogeneity conditions. The LMD criterion penalizes both mean mispricing and the cross-sectional dispersion of unit-level pricing errors, whereas the aggregated-moment criterion penalizes mean mispricing alone. Thus, even abstracting from weighting choices, the two sample objectives generally differ and need not share the same pseudo-true parameter under misspecification. This distinction is reinforced in Korteweg and Nagel (2016), where the private-equity pricing moment is excluded from the estimation criterion itself. Finally, the two approaches differ less in whether a large-time-span dimension is needed than in how that dimension is formalized. Korteweg and Nagel (2016) require sufficiently many nonoverlapping time windows in an aggregated-moment GMM setting. Our framework lets the number of vintage years grow in an increasing-domain random-field setting with bounded within-vintage counts and bounded fund lifetimes.

3.5.3 Positioning of the present estimator

The contribution of the present estimator is therefore not that it dominates prior approaches in every dimension. Rather, its advantage is methodological fit for the current application: it combines a transparent nonlinear least-squares objective with explicit control over the discounting horizon, accommodates pooled non-traded cash flows at the unit level, and admits an increasing-domain asymptotic framework that maps naturally to vintage-year variation. Table 3 summarizes the most prominent distinctions between the three approaches.

	Driessen et al. (2012)	Korteweg and Nagel (2016)	Our approach
Estimator type	Cross-sectional NLS (one-step GMM)	Time-series-like GMM	Nonlinear Least-Mean-Distance
Cash flows priced	PE fund cash flows	Public replicating strategies	PE fund cash flows
Discounting dates	Fund inception only	Fund inception only	Flexible via $\mathcal{T}_i$
Asymptotics	Infill ($n_i \to \infty$, N fixed)	$V \to \infty$	Increasing domain ($V \to \infty$, n_v bounded)
Inference	Bootstrap	Spatial HAC	Spatial HAC
Cross-sectional unit	Vintage year portfolio	Single fund	Both (tested empirically)
SDF specification	Linear factor model	Exponentially affine	Both (tested empirically)

Table 3: Comparison of SDF estimation frameworks for private equity.

4 Implementation choices and simulation

In this section and the next, we move from the general methodology to our empirical application. First, we describe the data and establish our sample restriction based on terminal cash-flow measurement concerns. Second, we detail the empirical implementation choices that anchor the general estimator from Section 3.2. Third, we use a simulation study to validate and calibrate these choices. Finally, Section 5 presents the resulting empirical estimates for Buyout and Venture Capital funds.

4.1 Private data

We use the Preqin cash flow dataset as of 14th April 2022, which is well known in the academic private equity literature (Harris et al., 2014; Brown et al., 2015; Korteweg and Nagel, 2016; Ang et al., 2018). To keep the empirical analysis clear and concise, we filter the dataset for Buyout (BO) and Venture Capital (VC) funds. We only keep funds with fund currency US Dollar (USD) and region equal to North America. In rare instances, the Preqin taxonomy assigns geographic labels that are

inconsistent with the North America region filter (e.g., a fund classified under a sub-regional or country-level tag that nonetheless passes the broader filter). Because all retained funds are denominated in USD, these classification inconsistencies do not introduce foreign-exchange contamination.

For this subset of BO and VC funds, we extract all (i) equal-weighted (hereafter EW) and (ii) value-weighted (hereafter FW, for fund-size-weighted) cash flow series (cf. Figures 2 and 3). In the overall dataset, the minimum vintage year is 1982, and the maximum is 2021. We initially retain all vintage years, including recent ones. Section 4.3 assesses the impact of unseasoned fund Net Asset Values (NAVs) on our empirical estimation results. On this basis, we decide how many recent vintage years to remove to reduce reliance on unrealized NAVs. For the vintage years we keep, all liquidated and non-liquidated funds enter our estimation. For non-liquidated funds, we treat the latest NAV as final cash flow.

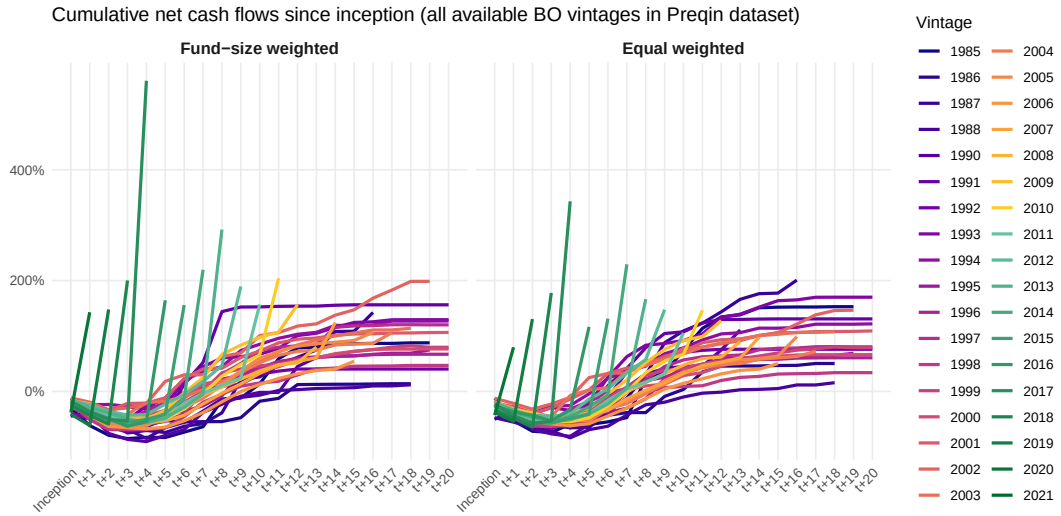

Figure 2: **Cumulative cash flows:** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 4.3, these vintages are excluded from the empirical estimates in Section 5 to limit reliance on unrealized NAVs.

Buyout sample. The Buyout (North America) sample contains 933 distinct funds spanning 36 vintage years. The vintage year distribution is as follows: 1985 (2), 1986 (3), 1987 (5), 1988 (6), 1990 (2), 1991 (3), 1992 (6), 1993 (5), 1994 (14), 1995 (11), 1996 (13), 1997 (14), 1998 (32), 1999 (27), 2000 (34), 2001 (21), 2002 (14), 2003 (21), 2004 (24), 2005 (44), 2006 (50), 2007 (49), 2008 (45), 2009 (19), 2010 (31), 2011 (25), 2012 (39), 2013 (36), 2014 (44), 2015 (35), 2016 (57), 2017 (35), 2018 (45), 2019 (65), 2020 (34), 2021 (23). The region distribution is as follows: Americas (4), Asia (16), Canada (1), Central and East Europe (2), Europe (6), Japan (1), Midwest (2), North America (461), South America (2), Southeast (1),

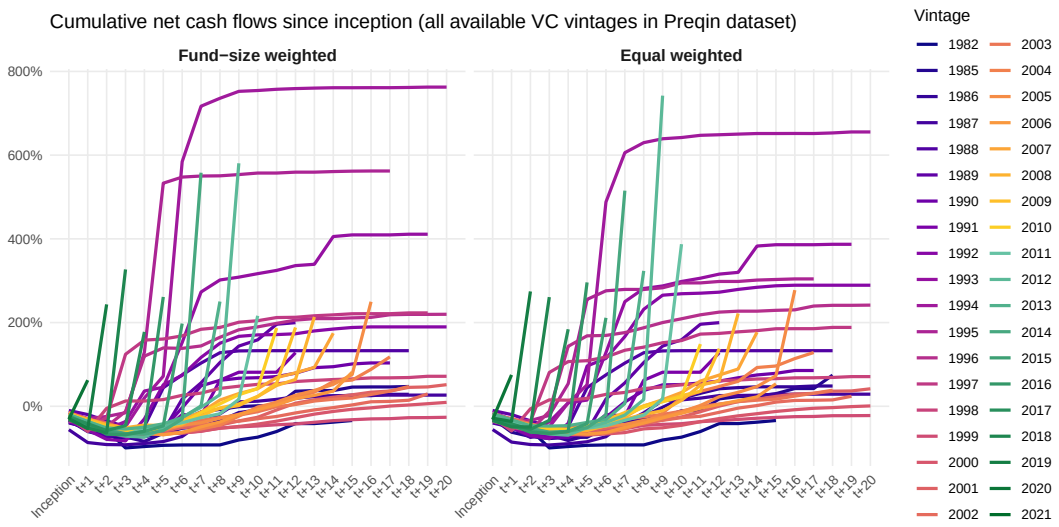

Figure 3: **Cumulative cash flows:** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 4.3, these vintages are excluded from the empirical estimates in Section 5 to limit reliance on unrealized NAVs.

Southwest (1), US (433), West Europe (1). The strategy distribution is as follows: Buyout (933). The fund currency distribution is as follows: USD (933).

Venture Capital sample. The Venture Capital (North America) sample contains 968 distinct funds spanning 38 vintage years. The vintage year distribution is as follows: 1982 (1), 1985 (3), 1986 (4), 1987 (2), 1988 (1), 1989 (1), 1990 (4), 1991 (1), 1992 (7), 1993 (6), 1994 (6), 1995 (8), 1996 (14), 1997 (16), 1998 (28), 1999 (44), 2000 (70), 2001 (45), 2002 (28), 2003 (18), 2004 (31), 2005 (34), 2006 (44), 2007 (48), 2008 (38), 2009 (18), 2010 (18), 2011 (23), 2012 (21), 2013 (32), 2014 (33), 2015 (37), 2016 (40), 2017 (50), 2018 (52), 2019 (64), 2020 (45), 2021 (33). The region distribution is as follows: Asia (10), Australasia (1), Canada (1), China (10), Europe (3), India (2), Mexico (1), Midwest (2), North America (267), Northeast (2), Southwest (1), Thailand (1), US (664), West (1). The strategy distribution is as follows: Early Stage (274), Early Stage: Seed (47), Early Stage: Start-up (26), Expansion / Late Stage (127), Venture (General) (472), Venture Debt (22). The fund currency distribution is as follows: USD (968).

4.2 SDF models and public data

We test a simple linear SDF model similar to Driessen et al. (2012)

$$\Psi^{\mathrm{SL}}_{\tau,t}(\theta) = \frac{\prod_{h=0}^{\tau} \ (1 + \alpha + r_{f,h} + \sum_m \ \beta_m \ F_{m,h})}{\prod_{h=0}^{t} \ (1 + \alpha + r_{f,h} + \sum_m \ \beta_m \ F_{m,h})} \tag{25}$$

and an exponential affine SDF model adapted from Korteweg and Nagel (2016)

$$\Psi_{\tau,t}^{\mathrm{EA}}(\theta) = \exp\left[-\sum_{h=\tau+1}^{t}\left(\alpha + \log(1+r_{f,h}) + \sum_{m\in M}\beta_m \cdot \log(1+F_{m,h})\right)\right] \tag{26}$$

with monthly (arithmetic) risk-free return r_f, monthly (arithmetic) zero-net-investment portfolio returns F_m, and parameter vector $\theta = (\alpha, \beta)$. Throughout both the simulation and the empirical application, all factor returns and fund cash flows are observed at monthly frequency.

The public market factors that enter our SDF draw on the US dataset of the recently popularized q^5 investment factor model sourced from `http://global-q.org/factors.html` (Hou et al., 2015, 2021). Their five-factor model includes the market excess return (MKT), a size factor (ME), an investment factor (IA), a return on equity factor (ROE), and an expected growth factor (EG). The maximum date in the q^5 investment factor dataset is 2024-12-31.

To avoid overfitting, we restrict our analysis to six simple SDF models that contain {MKT} alone or {MKT} plus {ME or IA or ROE or EG or Alpha}. In Equation 14, we use the quadratic loss function $L(x) = x^2$.

Remark 6 (Alpha Bounds)**.** *In our optimization procedure, we limit the Alpha factor α to the range of $[-2\%, +2\%]$ per month to avoid the "exploding alpha" issue documented in (Driessen et al., 2012, Footnote 8): "Note that one needs to put an upper bound on α when performing the optimization in equation (3). This is to exclude the possibility that α tends to infinity." In this degenerate scenario, an infinite alpha can almost perfectly price any cash flow stream (except its first component), rendering the beta risk factors irrelevant.*

4.3 The impact of unseasoned funds and reported NAVs

Figure 4 presents the vintage-year cutoff sensitivity analysis separately for BO and VC under fund-value weighting (FW). A clear upward-drift pattern is present in both sub-samples: as more recent, unseasoned vintages are included, the estimated market beta increases monotonically. For BO under value weighting, $\hat{\beta}_{\mathrm{MKT}}$ at the 0-year Horizon (see Remark 7) rises from approximately 1.23 with a 2011 vintage cutoff to roughly 2.42 with the full sample (2021). Notably, most of the increase is concentrated in vintages after 2016, where $\hat{\beta}_{\mathrm{MKT}}$ jumps sharply from 1.53 (2016 cutoff) to 2.34 (2017 cutoff), reflecting the disproportionate influence of young, unrealized funds. For VC, the drift is similarly dramatic: the FW estimate at the 0-year Horizon climbs from approximately 1.57 (2011 cutoff) to 2.33 with the full sample. Discounting all final NAVs by 50% substantially flattens the upward trend: the BO FW estimate range compresses to 1.09–1.65, and the VC FW range to 1.41–1.54, confirming that the drift is attributable to the outsized mechanical reliance on subjectively reported Net Asset Values for young funds. These findings reinforce the sample restriction to the 2010 maximum vintage year applied throughout the remainder of this section.

Remark 7 (Definition of Horizon)**.** *Throughout the empirical section, we use the term* Horizon *H to denote the maximum compounding horizon in years used in*

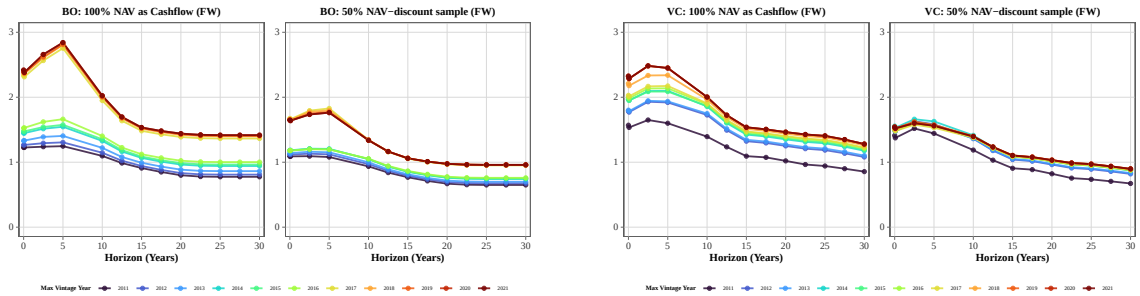

Figure 4: **Sensitivity analysis (MKT factor):** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

the set of discounting dates $\mathcal{T}$ (cf. Equation 13). Formally, starting from the fund inception date t_i^0, estimating a model with a Horizon H implies that the estimator minimizes the average pricing error over the set of discounting dates $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, formally denoted as $\bar{\epsilon}_i^{(H)}$. Note that the Horizon has as upper bound the max cash flow date per fund or portfolio in our empirical application. This Horizon in years serves as x-axis in all subsequent empirical and simulation charts.

4.4 NAV discount sensitivity analysis

To further quantify the sensitivity of factor loading estimates to terminal Net Asset Value treatment, we conduct a systematic NAV discount analysis on the seasoned sample (maximum vintage year 2010). Figure 5 reports single-factor MKT estimates under five NAV treatments: the baseline (full reported NAV as final cash flow), 25%, 50%, and 75% constant discounts applied to all final NAVs, and a complete NAV write-off (100% discount).

For BO, the estimated market beta decreases monotonically with the NAV discount. At the 0-year Horizon under EW, $\hat{\beta}_{\text{MKT}}$ drops from 1.26 (no discount) to 1.19 (25% discount), 1.12 (50%), 1.03 (75%), and 0.94 (100% write-off); under FW, the corresponding range is 1.20 to 0.98. At the 15-year Horizon, the EW estimate ranges from 0.81 (no discount) to 0.55 (100% write-off), while the FW estimate spans 0.84 to 0.62. The qualitative conclusion is robust across all NAV treatments: the single-factor market model remains economically plausible and stable.

For VC under EW, the pattern is analogous but the sensitivity is more muted in percentage terms due to the higher baseline beta: the 0-year estimate declines from 1.88 (no discount) to 1.84 (25%), 1.80 (50%), 1.77 (75%), and 1.74 (100%); at the 15-year Horizon, from 1.30 (no discount) to 1.09 (100% write-off). Under FW, the sensitivity is more pronounced, particularly at longer Horizons: the 0-year estimate ranges from 1.59 to 1.34, while the 15-year estimate declines from 1.07 to 0.73.

This analysis demonstrates that terminal NAV treatment has a material but

bounded effect on estimated factor loadings. The monotonic relationship confirms the intuition that reported NAVs mechanically inflate estimated betas: funds whose final cash flow is a subjective valuation rather than a realized exit price appear more systematically risky because their NAVs partially co-move with public market indices. Importantly, even under the extreme assumption of a complete NAV write-off, the first qualitative conclusions of this subsection can be summarized as follows: BO maintains a market beta near or below unity, while VC retains a substantially higher systematic exposure. We do not impose a model-based residual-value adjustment, as this would constitute a separate research agenda (Jenkinson et al., 2020; Brown et al., 2023). Instead, the range spanned by the five NAV treatments provides a transparent envelope of estimation uncertainty attributable to terminal value measurement.

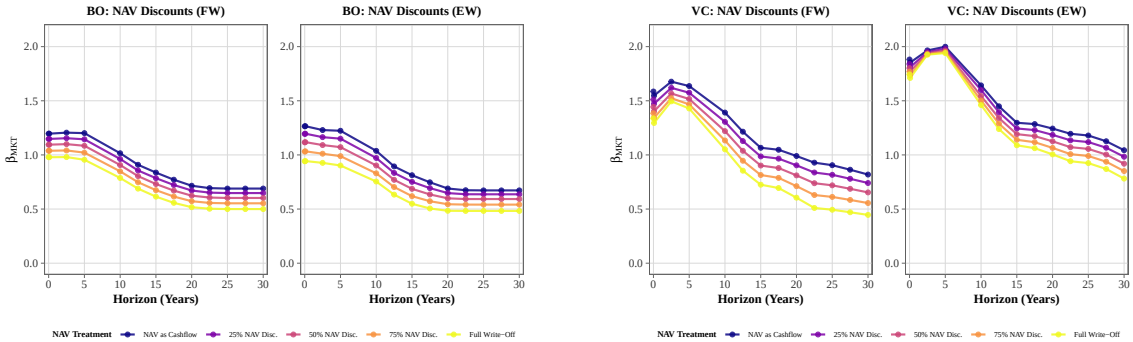

Figure 5: **NAV discount sensitivity (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off.

4.5 Simulation study

Our Monte Carlo experiments examine the bias and variance of our estimation methodology in finite samples.[2] Specifically, we investigate five key dimensions of the estimation framework: (i) the finite-sample efficiency gains of forming vintage-year portfolios versus using individual funds; (ii) the tradeoff between expanding the time-series dimension (number of vintage years) and the cross-sectional dimension (funds per vintage); (iii) the robustness of the estimates to the functional specification of the SDF (simple linear vs. exponential affine); (iv) the behavior of multivariate identification under idiosyncratic noise using two-factor models; and (v) as our main overarching topic, the optimal choice of the maximum compounding Horizon (i.e., the set of discounting dates $\mathcal{T}$).

[2]As with any simulation study, it investigates the ability to identify the assumed data generating process rather than validating the corresponding SDF model itself.

We use historical q-investment factors from 1986 to 2005 and simulate 20 funds for each of these 20 vintage years. This means we condition our simulation analysis on realized q-investment factors paths (see the Macro-Conditioning Kernel discussion in Section 3.1). Each fund contains 15 deals with equal investment amounts and exactly one divestment cash flow. Deals are entered within the first five years of the fund lifetime following a discrete uniform distribution and afterward held between one to ten years again uniformly distributed. The deal returns are generated by the simple linear or exponential affine SDF models described in Equations 25 and 26. In the base case, we just use the MKT factor with $\beta_{\text{MKT}} = 1$ and in each month, add a normal i.i.d. error term with monthly standard deviation $\sigma = 0.2$ (annualized variance $12 \cdot 0.2^2 = 0.48$, corresponding to an annualized volatility of $0.2\sqrt{12} \approx 69\%$) and zero mean. In the exponential affine case, we adjust the lognormally distributed error mean to zero by subtracting $0.5\sigma^2$. If a negative return exceeds -100%, the company defaults with a zero exit cash flow. In contrast, the error term in the simulations of Driessen et al. (2012) is more well-behaved as it follows a shifted lognormal distribution that, even with arbitrarily high error term variance, just allows for returns below say -99%, if the market return is close to its lower bound (see equation 9 in their online appendix). We specifically analyze the remaining divergence from their simulation design in Appendix B.1. In our simulation scenarios, we evaluate the estimator across a full grid of compounding Horizons, where the maximum month in the set of discounting dates $\mathcal{T}$ ranges from 0 up to 30 years (0-360 months). To assess our estimator's bias and variance, we simulate 1000 test scenarios for both vintage-year and single-fund portfolios.

Cross-sectional unit i: As anticipated in Subsection 3.3, vintage year portfolios demonstrate lower bias and variance than individual funds. Under the simple linear SDF and a 15-year Horizon (maximum month 180), the mean (standard deviation) of the coefficient estimate $\hat{\beta}_{\text{MKT}}$ is 0.966 (0.188) for vintage year portfolios, compared to 0.887 (0.359) for individual funds. The superiority of the portfolio approach is especially striking when evaluating the entire Horizon curve in Figure 6. Estimates for individual funds are highly unstable, exhibiting severe upward bias at short Horizons ($\hat{\beta}_{\text{MKT}} \approx 1.5$) and deteriorating to negative values at long Horizons. In contrast, vintage year portfolios yield substantially more stable estimates that remain within a narrow, economically plausible range around the true value of 1.0. The portfolio approach consistently achieves a lower standard deviation across all simulated Horizons, confirming that grouping funds effectively mitigates idiosyncratic noise.

This trade-off between bias/variance reduction and sample size has important implications. While forming vintage year portfolios substantially improves the accuracy of the estimates, it dramatically reduces the number of cross-sectional units, thereby weakening the relevance of asymptotic approximations. This constraint likely explains the choice of Korteweg and Nagel (2016) to rely on individual funds as cross-sectional units in their asymptotic SHAC framework, enabling them to exploit a larger sample size for smaller standard errors.

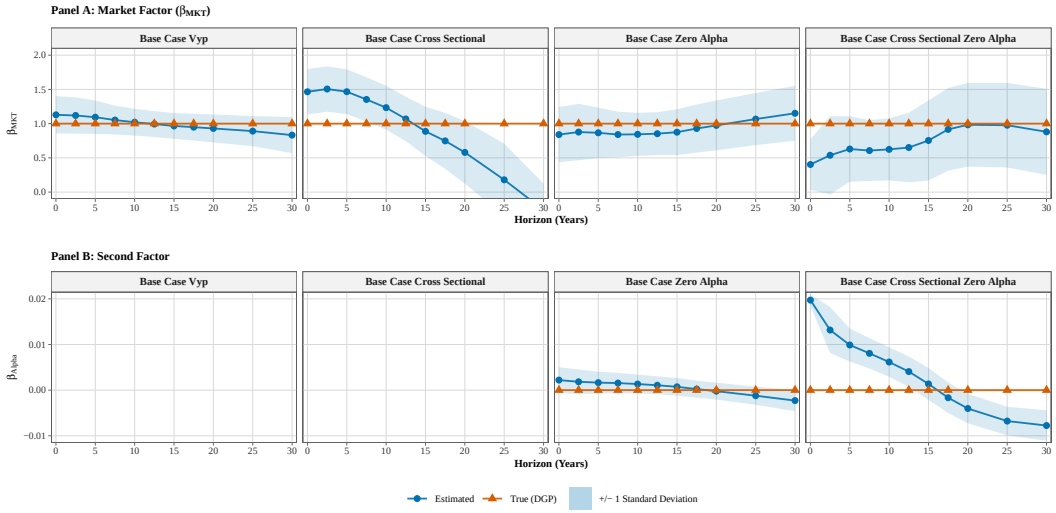

Figure 6: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Single-fund vs. vintage-year portfolios.

Varying vintages V and portfolio sizes n/V: A fundamental question in private equity panel data econometrics is whether precision is best improved by extending the time-series dimension (V, number of vintage years) or the cross-sectional dimension (n/V, number of funds per vintage portfolio). To assess this tradeoff, we compare five data configurations (depicted in Figures 7 and 8): (1) our base case for this experiment ($V = 20$, $n/V = 40$); (2) an extended vintage sample with small portfolios ($V = 39$, $n/V = 10$); (3) an extended vintage sample with medium portfolios ($V = 39$, $n/V = 20$); (4) an early-vintage sub-sample ($V = 10$, years 1986–1995, $n/V = 20$); and (5) a late-vintage sub-sample ($V = 10$, years 1996–2005, $n/V = 20$).

In our simulation design, increasing the number of cross-sectional units per portfolio (n/V) is substantially more effective at suppressing estimator variance than adding more vintage years (V). For the single-factor model at the 15-year Horizon, transitioning from 10 funds to 20 funds per vintage over a 39-year period (scenarios 2 vs. 3) reduces the standard deviation of $\hat{\beta}_{\text{MKT}}$ by roughly 22%, from 0.218 to 0.169. In contrast, shrinking the time-series dimension from 39 years to just 10 years (while holding $n/V = 20$ constant) elevates the variance only moderately for the early-vintage sub-sample ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.194$ for 1986–1995), though the late-vintage sub-sample (1996–2005) exhibits considerably higher variance ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.432$), reflecting the sensitivity to the realized macro environment. This hierarchy of variance reduction is even more pronounced in the two-factor setting, where idiosyncratic cash flow noise severely impairs multivariate identification: increasing n/V aggressively dampens the spurious α variance. Consequently, our findings provide an empirical rationale for the finite-sample superiority of the Driessen et al. (2012) infill asymptotics (fixing V, letting $n_v \to \infty$) over approaches that rely purely on time-series aggregation.

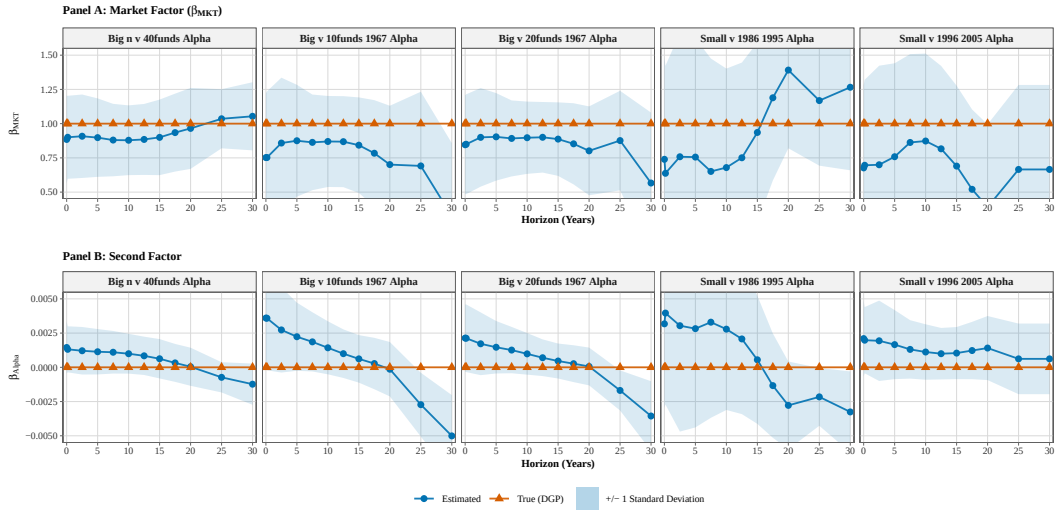

Figure 7: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying vintages V and portfolio sizes n/V.

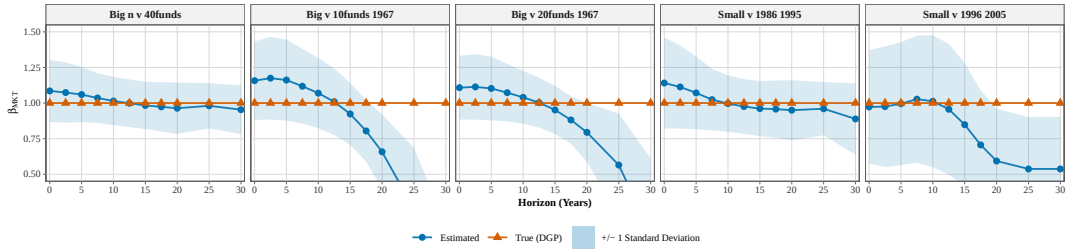

Figure 8: **Simulation ($\beta_{\mathbf{MKT}}$):** Varying vintages V and portfolio sizes n/V.

SDF model Ψ: In Figure 9, we compare the simple linear SDF model against the exponential affine specification. For the single-factor model, the bias is minimized at the 12.5-year Horizon (150 months), where both functional forms perform almost identically and yield estimates extremely close to the true value of 1.0: the exponential affine yields a mean market beta of 0.999 (0.182), matching the simple linear's 0.994 (0.189). For the two-factor (MKT+Alpha) model at this same 12.5-year Horizon, both functional forms again yield nearly identical parameter results, though with an attenuated market beta ($\hat{\beta}_{\text{MKT}} \approx 0.85$) and a minor spurious positive alpha ($\hat{\alpha} \approx 0.001$). In the two-factor case, the market beta bias is actually minimized later, around the 20-year Horizon ($\hat{\beta}_{\text{MKT}} \approx 0.97$, $\hat{\alpha} \approx -0.0002$).

However, a salient difference emerges at extremely short compounding Horizons. At a 1-month Horizon (resembling an NPV-only discounting framework), the exponential affine formulation deteriorates much more severely in both models, manifesting a larger upward bias (e.g., in the single-factor case, $\hat{\beta}_{\text{MKT}} = 1.21$ vs. 1.13) and elevated variance (0.39 vs. 0.27) compared to the linear model. Overall, the perceived superiority of exponential affine SDFs (Gourieroux and Monfort, 2007;

Bertholon et al., 2008) does not translate into empirical finite-sample advantages in our framework. The simple linear model actually appears slightly better behaved.

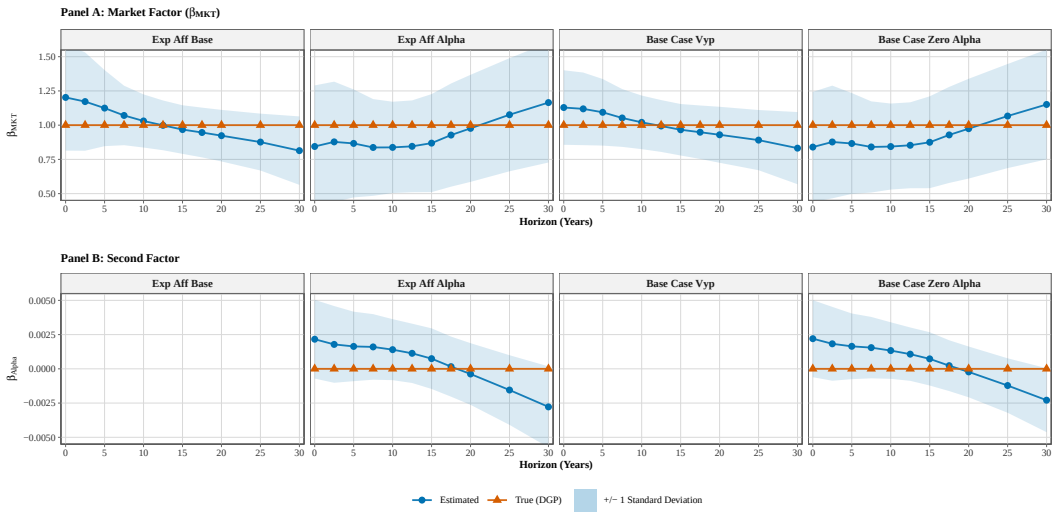

Figure 9: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Exponential affine vs. linear models.

Two-factor models (q^5 factors): Figure 10 reports simulation results for two-factor models that pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG), with true loadings $\beta_{\text{MKT}} = 1$ and $\beta_{\text{2nd}} = 0.5$. A common pattern emerges across all four specifications: the second-factor loading is substantially overestimated at short Horizons and attenuated toward (or beyond) the true value at long Horizons, while the market factor absorbs a compensating bias in the opposite direction. Estimation variance is markedly larger than in the single-factor case, reflecting the collinearity between correlated factors in a limited cross-section of vintage year portfolios. Table 4 summarizes the key estimates at the shortest Horizon and at 15 years, which is approximately the MSE-minimizing Horizon for most models; we highlight the most notable factor-specific features below.

The ME model is the only specification in which $\hat{\beta}_{\text{2nd}}$ undergoes a sign change (at roughly 17.5 years), declining from 0.86 to -0.02 over the full Horizon range, a severe attenuation that would qualitatively alter economic conclusions. The IA model exhibits a large second-factor bias at short Horizons ($\hat{\beta}_{\text{IA}} \approx 1.06$ vs. truth 0.5) but both loadings converge close to their true values near the 17.5-year Horizon ($\hat{\beta}_{\text{MKT}} = 0.97$, $\hat{\beta}_{\text{IA}} = 0.52$). The EG model exhibits the largest overall estimation variance ($\sigma(\hat{\beta}_{\text{EG}}) \approx 1.5$ at the shortest Horizon) but achieves a reasonably accurate joint estimate around 15–17.5 years ($\hat{\beta}_{\text{MKT}} = 0.91$–$1.02$, $\hat{\beta}_{\text{EG}} = 0.47$–$0.56$). The ROE model is the exception: both loadings remain persistently biased at all Horizons, with $\hat{\beta}_{\text{MKT}}$ never exceeding 0.91 and $\hat{\beta}_{\text{ROE}}$ never falling below 0.62. These results suggest that empirical researchers should exercise caution when interpreting two-factor SDF estimates for private equity, and that the MSE-minimizing Horizon is model-dependent, typically falling in the 12.5–17.5-year range.

Model	$\hat{\beta}_{\text{MKT}}$		$\hat{\beta}_{\text{2nd}}$		$\sigma(\hat{\beta}_{\text{MKT}})$		$\sigma(\hat{\beta}_{\text{2nd}})$	
	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr
ME + MKT	1.15	0.97	0.87	0.59	0.28	0.21	0.46	0.40
IA + MKT	0.97	0.92	1.06	0.64	0.34	0.30	0.74	0.53
ROE + MKT	0.88	0.83	0.98	0.69	0.43	0.49	0.61	0.49
EG + MKT	0.87	0.91	0.99	0.56	0.61	0.45	1.53	0.31

True values: $\beta_{\text{MKT}} = 1$, $\beta_{\text{2nd}} = 0.5$.

Table 4: Two-factor simulation summary: mean estimates and standard deviations at the shortest (0 yr) and the 15-year Horizon.

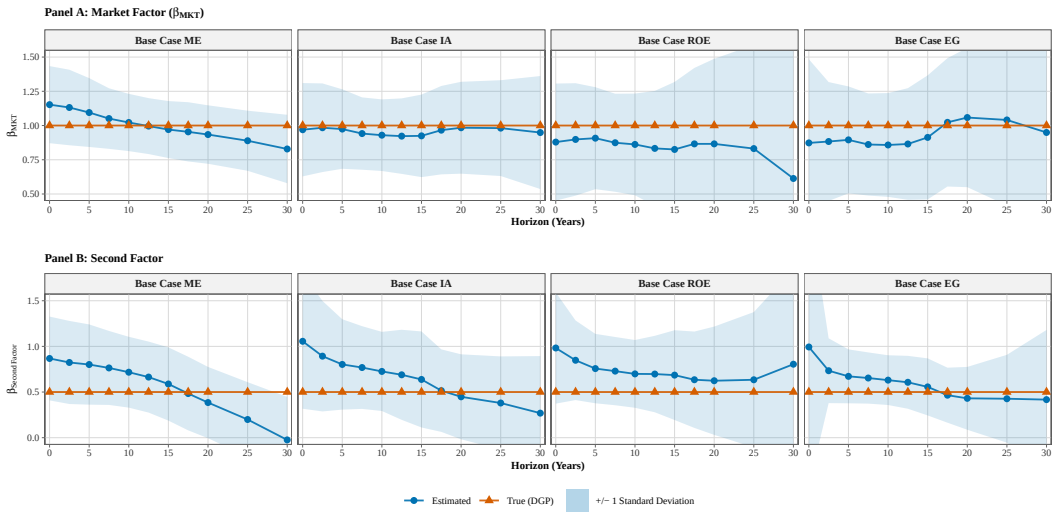

Figure 10: **Simulation (q^5 factors):** Two-factor models using VYP.

Varying fund lifetimes: Figure 11 illustrates how the optimal Horizon shifts in response to abbreviated fund lifetimes. As anticipated, the intercept of the alpha parameter decreases for shorter fund lifetimes relative to our baseline scenario of a five-year investment period and a ten-year maximum holding period. Specifically, for the shortest examined fund lifetime of two years (comprising a one-year investment period and a one-year maximum holding period), the beta bias is minimized at a 12.5-year Horizon, while the alpha bias is minimized at a 5-year Horizon. Similarly, for a fund with a two-year investment period and a two-year maximum holding period, the minimum beta bias occurs at a 15-year Horizon, and the minimum alpha bias at a 7.5-year Horizon. Consequently, these results demonstrate that for funds with short lifespans, the optimal Horizon extends significantly beyond the fund's actual lifetime.

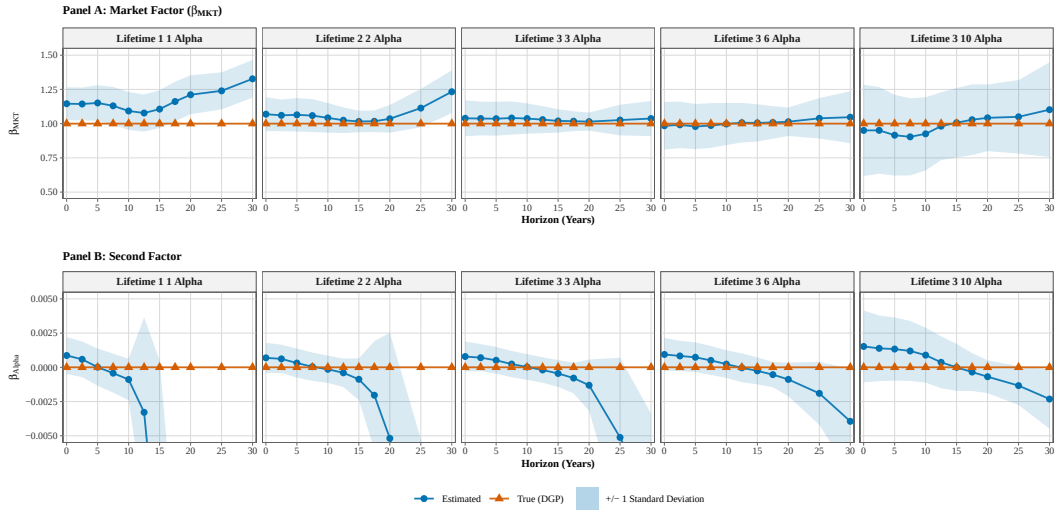

Figure 11: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying fund lifetimes using VYP with the following yearly investment-period/holding-period pairs: 1/1, 2/2, 3/3, 3/6, and 3/10 (from left to right).

Size of set $\mathcal{T}$ (Maximum Compounding Horizon): Our most critical simulation finding relates to the optimal choice of the maximum compounding Horizon, i.e., the size of the set of discounting dates $\mathcal{T}$. From a purely theoretical perspective (cf. Section 3.1), discounting all cash flows exclusively to the fund inception date, corresponding to a zero-month or NPV-only Horizon in our notation, should yield unconditionally unbiased estimates under the deal-level exact-pricing restriction (Equation (2)). This minimal choice for $\mathcal{T}$ corresponds exactly to the established Driessen et al. (2012) framework.

However, our diverse simulation results reveal a puzzling disconnect between asymptotic theory and finite-sample performance: extending the set of discounting dates deep into the fund lifecycle drastically improves the estimator's precision. Across almost all model specifications (see Figures 10 and 9), bias is minimized not at the theoretical 0-year Horizon, but rather around a 12.5 to 15-year compounding Horizon. At extremely short Horizons, idiosyncratic cash flow noise interacts non-linearly with the estimation procedure, inducing severe bias in factor loadings. By averaging pricing errors over multiple future dates, effectively incorporating the Timing Risk Premium, our approach acts as finite-sample stabilization device.

Interestingly, this "variance reduction technique" improves the mean estimate far more than it reduces standard errors. While estimator variance traces a shallow U-shape with a minimum near the 15-year Horizon (see Table 4) for fund with a lifetime of 15 years, this reduction is secondary to the striking stabilization of the parameter bias. However, our simulations for shorter fund lifetimes indicate that determining the optimal horizon requires tailored simulation designs rather than a simple heuristic assuming the optimal horizon always coincides with the fund's life. We further analyze these puzzling results from a theoretical perspective in Appendix A, detailing the finite-sample stabilization under $\mathcal{T}$-averaging and the role of the

Timing Risk Premium.

Simulation conclusion: To conclude, our simulation study rationalizes two key practices from the Driessen et al. (2012) paper: (i) vintage-year portfolio formation helps to improve estimator precision, and (ii) increasing the number of funds per vintage seems more effective in controlling estimator variance than increasing the number of vintages (for i.i.d. error terms). Our second finding provides an empirical rationale for the choice of Driessen et al. (2012) to employ infill asymptotics that let the number of underlying funds per vintage tend to infinity. However, our correctly specified scenarios cannot support the assumption of Korteweg and Nagel (2016) that (iii) the exponential affine SDF is superior to the simple linear SDF in a multi-period framework; in fact, their finite-sample bias and variance profiles are remarkably similar. Our simulation study also suggests that (iv) averaging pricing errors over multiple dates substantially reduces the severe short-Horizon bias inherent to the original procedure of Driessen et al. (2012), which discounts all cash flows exclusively to the fund inception date. Ultimately, selecting a set of compounding dates $\mathcal{T}$ that spans a significant portion of the fund lifecycle appears to decrease finite-sample bias (and to a lesser extent, variance) more effectively than all other measures combined. Appendix A explains the theoretical foundation of this observation by showing how NFV-induced timing terms can shift the population target while still improving finite-sample performance.

Accordingly, in the empirical application we use vintage-year portfolios, the simple linear SDF, emphasize one-factor MKT specifications, treat two-factor results as exploratory, and focus on Horizons around 12.5–15 years while also reporting sensitivity over a broader grid.

4.6 Role of asymptotic inference

To assess the parameter significance in the empirical application, we compute the asymptotic standard errors as outlined in Subsection 3.4. For the Bartlett kernel's bandwidth $b_n = D$ we select 12 years, i.e., funds with absolute vintage year differences larger than 12 years are assumed to be independent. The asymptotic results provide the benchmark large-sample approximation.

At the same time, private-equity samples remain small in the vintage-year and cross-sectional dimensions, and model choice is often data-driven. For that reason, the empirical analysis complements asymptotic inference with out-of-sample validation and stability diagnostics. To account for the dependency between funds from adjacent vintage years caused by overlapping fund cash flows, we draw on hv-block cross-validation (Racine, 2000). As Racine (2000) demonstrates, hv-block cross-validation is a consistent model-selection criterion for dependent data: the h-block gap between training and validation sets ensures approximate independence, while the v-block provides a sufficiently large validation sample within each fold. In our implementation, each cross-validation fold partitions the data into three disjoint groups: (i) a *validation set* (v-block) comprising funds from three consecutive vintage years (e.g., 2000, 2001, 2002), on which the out-of-sample pricing error is evaluated; (ii) a *buffer zone* (h-block) of three vintage years on each side of the

v-block (e.g., 1997–1999 and 2003–2005), which is discarded entirely to break the dependence between training and validation data caused by overlapping fund cash flows; and (iii) an *estimation set* consisting of all remaining vintage years (e.g., 1985–1996 and 2006–end), on which the model parameters are estimated. This three-way split is repeated across rolling vintage-year windows, as detailed in Table 5. The number of active folds adapts dynamically to the selected maximum vintage year cutoff in any given estimation run. For instance, when estimating the full sample through 2021, we apply ten folds; when estimating a heavily seasoned sample truncated at 2010 to avoid reliance on subjective interim NAVs, the framework naturally truncates to seven active folds.

Remark 8 (Cross-validation as stability diagnostic)**.** *The hv-block cross-validation quantities reported throughout this paper serve as* ***stability diagnostics and model-selection criteria*** *rather than inferential standard errors. They summarize the fold-to-fold variation in parameter estimates under dependence-reducing sample splits and are informative about model sensitivity and effective parameter curvature, but they do not provide a valid null distribution for the full-sample estimator. For formal statistical inference (standard errors and hypothesis tests), we rely on the SHAC-based asymptotic framework of Section 3.4. In practice, hv-block cross-validation complements the asymptotic analysis by (i) diagnosing model stability and out-of-sample performance without relying on asymptotics and (ii) flagging over-parameterized specifications that appear well-identified in-sample but collapse out-of-sample.*

training.before	h-block.before	v-block	h-block.after	training.after
estimation	remove	validation	remove	estimation
start-1984	1985,1986,1987	1988,1989,1990	1991,1992,1993	1994-end
start-1987	1988,1989,1990	1991,1992,1993	1994,1995,1996	1997-end
start-1990	1991,1992,1993	1994,1995,1996	1997,1998,1999	2000-end
start-1993	1994,1995,1996	1997,1998,1999	2000,2001,2002	2003-end
start-1996	1997,1998,1999	2000,2001,2002	2003,2004,2005	2006-end
start-1999	2000,2001,2002	2003,2004,2005	2006,2007,2008	2009-end
start-2002	2003,2004,2005	2006,2007,2008	2009,2010,2011	2012-end
start-2005	2006,2007,2008	2009,2010,2011	2012,2013,2014	2015-end
start-2008	2009,2010,2011	2012,2013,2014	2015,2016,2017	2018-end
start-2011	2012,2013,2014	2015,2016,2017	2018,2019,2020	2021-end

Table 5: Partitions used for hv-block cross-validation. The number of active folds truncates dynamically based on the maximum vintage year cutoff of the specific estimation run.

5 Empirical results for BO and VC data

This section presents the empirical SDF estimates for Buyout (BO) and Venture Capital (VC) funds. As the simulation study in Subsection 4.5 confirms, we use vintage-year portfolios (VYPs) to estimate simple linear SDF models for the BO and VC sub-samples individually. We restrict the data to North American, USD-denominated funds from the Preqin dataset described in Subsection 4.1. This restriction ensures consistency between fund cash flows and the U.S. q-investment factors used throughout the analysis. Moreover, we only consider funds with a maximum vintage year of 2010, to reduce the reliance on unrealized NAVs (see Section 4.3.

We analyze BO and VC as two distinct asset classes with fundamentally different risk profiles and report both equal-weighted (EW) and value-weighted (FW) estimates across a grid of compounding Horizons. Our baseline specification is the single-factor MKT model estimated on vintage-year portfolios. Two-factor models that add an alpha intercept or a single q-investment factor serve as robustness and exploratory extensions.

For each model specification, we report asymptotic standard errors (SE_A) from the SHAC framework alongside hv-block cross-validation fold dispersions (SD_{CV}), defined as the standard deviation of parameter estimates across hold-out folds. Throughout the empirical analysis, t_A denotes the t-ratio based on asymptotic standard errors; SR_{CV} (stability ratio) denotes the ratio of the full-sample estimate to the cross-validation fold dispersion, i.e., $\mathrm{SR}_{CV} := \hat{\theta}/\mathrm{SD}_{CV}$. As discussed in Remark 8, SR_{CV} complements asymptotic inference as a stability diagnostic.

All R code and data (including the previous simulations) are available in an online repository: `https://github.com/quant-unit/Fundwise_SDF/tree/master/r_project`.

5.1 Baseline: single market-factor models

Tables 6–7 and Figure 12 report the single-factor MKT model for BO and VC across Horizons from 0 to 30 years. As anticipated by the simulation study, the market beta declines with the compounding Horizon for both asset classes.

For BO under EW, $\hat{\beta}_{\mathrm{MKT}}$ ranges from approximately 1.22–1.27 at short Horizons (0–5 years) and decreases to 0.67 at 25–30 years; under FW, the corresponding range is 1.20–1.21 down to 0.69. The divergence between asymptotic and cross-validation diagnostics persists: asymptotic t-ratios (t_A) are moderate at short Horizons (e.g., 1.10 under EW at 0 years) and decline further at longer Horizons, while the cross-validation stability ratios remain high throughout (SR_{CV} ranging from 3.75 to 10.65 under EW and from 2.63 to 8.68 under FW), indicating robust stability across data subsets.

For VC, the market beta is substantially higher across all Horizons, consistent with venture capital's greater sensitivity to systematic risk. Under EW, $\hat{\beta}_{\mathrm{MKT}}$ reaches approximately 1.97–2.00 at 2.5–5 years and converges toward 1.04–1.13 beyond 25 years; under FW, the corresponding range is 1.64–1.68 down to 0.82–0.86. Despite the higher parameter variability reflected in larger asymptotic standard errors (SE_A of 1.2–1.8), the cross-validation stability diagnostics indicate robust

estimates (SR_{CV} of 3.59–4.46 under EW; 3.54–7.01 under FW), although somewhat lower than the BO stability ratios due to the more heterogeneous VC fund universe. The converged long-horizon estimates (BO $\approx$ 0.67, VC $\approx$ 1.0–1.1 under EW) are economically plausible: BO funds, which invest in mature companies with stable cash flows, exhibit market exposure near or slightly below unity, while VC funds, which back early-stage ventures with highly uncertain outcomes, retain a substantially higher beta even at seasoned Horizons.

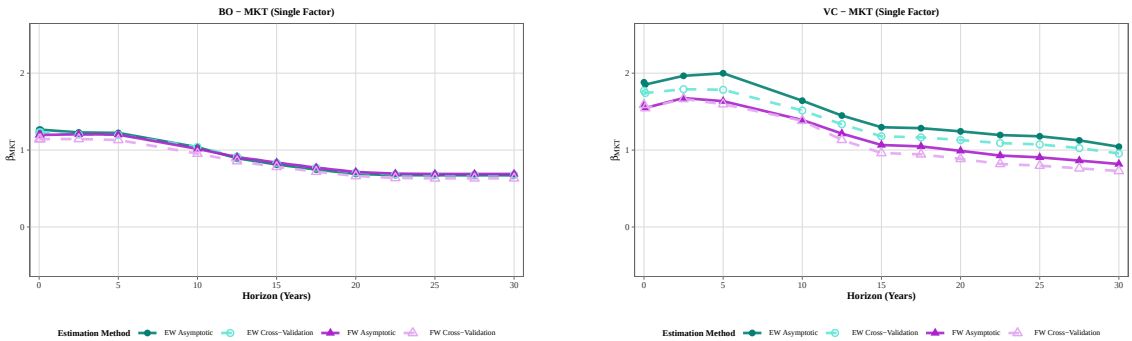

Figure 12: **Empirical (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

Table 6: Empirical Estimates: BO — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.264	1.154	0.123	1.10	10.00	1.197	1.170	0.139	1.02	8.23
0.1 yr	1.265	1.155	0.116	1.10	10.65	1.196	1.170	0.132	1.02	8.68
2.5 yr	1.229	0.975	0.133	1.26	9.07	1.206	1.175	0.170	1.03	6.73
5 yr	1.223	0.972	0.147	1.26	8.16	1.201	1.173	0.196	1.02	5.78
10 yr	1.038	1.642	0.189	0.63	5.52	1.015	1.806	0.228	0.56	4.19
12.5 yr	0.894	1.660	0.224	0.54	4.08	0.909	1.627	0.247	0.56	3.46
15 yr	0.812	1.511	0.223	0.54	3.75	0.837	1.798	0.246	0.47	3.18
17.5 yr	0.747	1.405	0.206	0.53	3.75	0.771	1.699	0.248	0.45	2.90
20 yr	0.690	2.001	0.163	0.34	4.36	0.715	1.737	0.245	0.41	2.71
22.5 yr	0.674	1.947	0.136	0.35	5.03	0.693	1.687	0.241	0.41	2.65
25 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
27.5 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
30 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

Table 7: Empirical Estimates: VC — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.881	1.765	0.402	1.07	4.40	1.586	1.645	0.359	0.96	4.42
0.1 yr	1.852	1.756	0.390	1.06	4.46	1.548	1.651	0.335	0.94	4.62
2.5 yr	1.965	1.803	0.451	1.09	3.97	1.677	1.646	0.453	1.02	3.67
5 yr	1.999	1.824	0.493	1.10	3.62	1.636	1.643	0.437	1.00	3.65
10 yr	1.642	1.273	0.364	1.29	4.15	1.391	1.783	0.318	0.78	4.35
12.5 yr	1.449	1.269	0.304	1.14	4.39	1.214	2.005	0.151	0.61	7.49
15 yr	1.297	1.215	0.264	1.07	4.46	1.066	6.793	0.137	0.16	7.01
17.5 yr	1.285	1.216	0.279	1.06	4.17	1.048	6.123	0.148	0.17	6.36
20 yr	1.243	1.220	0.283	1.02	4.00	0.990	4.655	0.159	0.21	5.55
22.5 yr	1.195	1.320	0.272	0.91	4.02	0.929	4.906	0.176	0.19	4.66
25 yr	1.179	1.314	0.272	0.90	3.95	0.905	4.420	0.183	0.20	4.36
27.5 yr	1.126	1.292	0.269	0.87	3.81	0.863	3.749	0.194	0.23	3.93
30 yr	1.043	1.600	0.266	0.65	3.59	0.819	3.220	0.206	0.25	3.54

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

5.2 Robustness: two-factor $\alpha + MKT$ models

Tables 8–9 and Figure 13 present the two-factor MKT + α model for BO and VC. The results show the pathological behavior anticipated by the simulation study: once an unconstrained alpha intercept is introduced, the estimation destabilizes and the MKT loading collapses.

For BO, the alpha term absorbs a substantial share of the explanatory power: under EW, $\hat{\alpha}$ ranges from 0.003 to 0.006 per month and the MKT coefficient oscillates erratically between 0.41 and 1.01, with no stable convergence across Horizons; under FW, $\hat{\alpha}$ is persistently around 0.005–0.008 per month while the MKT loading collapses to values between 0.10 and 0.32. The asymptotic standard errors are highly erratic, with several entries at or very near zero interspersed with values exceeding 25. Cross-validation fold dispersions for $\hat{\beta}_{\mathrm{MKT}}$ are large (0.09–0.49 under EW; 0.65–0.74 under FW), confirming that the collapsed point estimates are unstable across data subsets.

For VC, the two-factor model exhibits a distinctive and economically puzzling pattern. At short Horizons (0–0.1 years under EW), the alpha term reaches its upper bound ($\hat{\alpha} = 0.020$/month) and the MKT coefficient drops to approximately 0.47, resembling the BO and PE behavior. However, as the Horizon increases beyond 5 years, the MKT loading rises steadily to values exceeding 2.3, while the alpha estimate turns progressively negative, reaching -0.009/month beyond 25 years. This reversal is unique to VC and likely reflects the extreme idiosyncratic risk inherent to early-stage ventures: at long Horizons, the optimizer increasingly attributes the high compounded returns of surviving VC funds to market co-movement rather than alpha, while the negative alpha compensates for selection effects among the funds that underperformed. Under FW, the instability is somewhat attenuated ($\hat{\alpha}$

remains close to zero at short Horizons; $\hat{\beta}_{\text{MKT}} \approx 1.73$), but the overall fragility persists, with asymptotic standard errors exceeding 7–11 across most Horizons.

The BO and VC two-factor results therefore corroborate the conclusion drawn for PE: single-factor MKT models are the most statistically robust specification for private equity SDF estimation in our sample, given the limited cross-section of vintage-year portfolios.

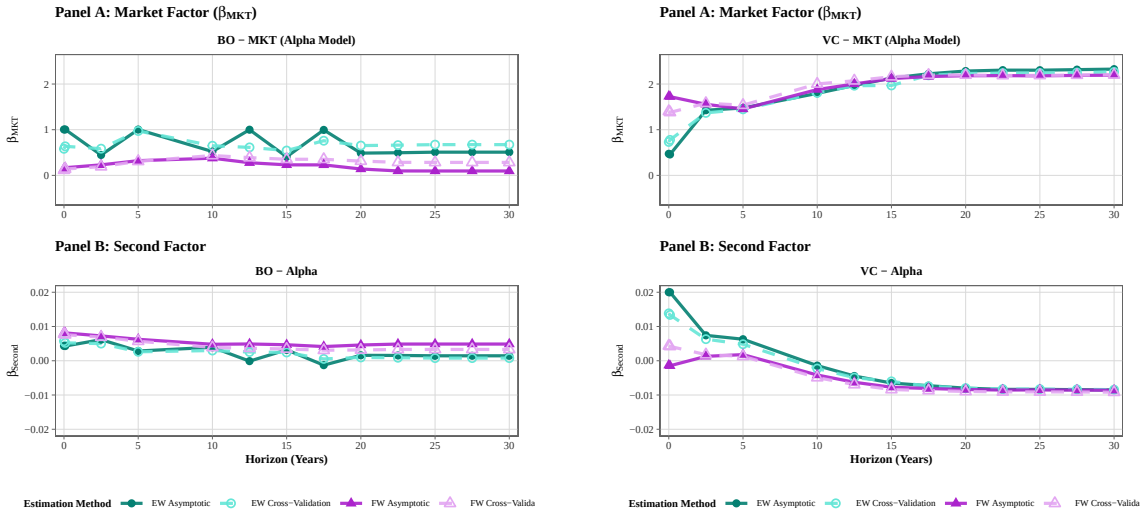

Figure 13: **Empirical ($MKT + \alpha$ model):** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

5.3 Two-factor q^5-factor models

We now pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG) in two-factor specifications estimated separately for BO and VC. Because the detailed estimation tables contain extensive Horizon-by-Horizon results, we do not reproduce them here and instead summarize the key patterns visible in Figures 14 and 15; the full tables are available in the accompanying online repository.

Buyout. Figure 14 shows a uniform pattern across the four non-market factors. At short Horizons (0–5 years), the second-factor loadings are positive and economically large: the ME loading reaches approximately 1.4–1.6 under EW (with stability ratios SR_{CV} of 3.0–3.8), the IA loading starts near 1.0 under EW ($\text{SR}_{CV} \approx 1.4$), ROE near 0.6 ($\text{SR}_{CV} \approx 0.9$), and EG near 0.5 ($\text{SR}_{CV} \approx 1.3$). As the Horizon increases, however, all four second-factor loadings decay toward zero and frequently change sign beyond 10–15 years. The ME factor retains a moderate positive loading even at long Horizons under FW ($\hat{\beta}_{\text{ME}} \approx 1.3$, $\text{SR}_{CV} \approx 2.1$), but under EW the loading drops below 1.0 after 17.5 years. The EG, IA, and ROE loadings all fluctuate around zero at intermediate and long Horizons, with cross-validation stability ratios well below conventional thresholds. The MKT loading in these two-factor models remains similar to its single-factor counterpart (ranging from roughly 1.0–1.1 at

Table 8: Empirical Estimates: BO — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	1.005	1.066	0.489	0.94	1.19	0.147	25.778	0.711	0.01	0.16
0.1 yr	1.005	1.069	0.430	0.94	1.48	0.168	24.142	0.716	0.01	0.19
2.5 yr	0.446	1.379	0.377	0.32	1.55	0.230	6.788	0.739	0.03	0.26
5 yr	1.003	2.078	0.087	0.48	11.22	0.322	6.558	0.707	0.05	0.43
10 yr	0.522	40.420	0.298	0.01	2.18	0.376	3.337	0.652	0.11	0.67
12.5 yr	1.000	3.644	0.386	0.27	1.59	0.277	5.044	0.651	0.05	0.60
15 yr	0.406	0.385	0.357	1.06	1.53	0.231	0.453	0.682	0.51	0.52
17.5 yr	0.995	2.953	0.364	0.34	2.08	0.230	11.801	0.707	0.02	0.50
20 yr	0.488	0.788	0.435	0.62	1.50	0.140	14.154	0.723	0.01	0.43
22.5 yr	0.495	0.779	0.451	0.64	1.47	0.097**	0.035	0.726	2.75	0.39
25 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
27.5 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
30 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
Panel B: Alpha (β_{Alpha})										
0 yr	0.004	0.006	0.004	0.74	1.54	0.008	0.192	0.005	0.04	1.50
0.1 yr	0.004	0.006	0.003	0.73	1.58	0.008	0.179	0.005	0.05	1.46
2.5 yr	0.006	0.014	0.003	0.45	1.83	0.007	0.055	0.005	0.13	1.30
5 yr	0.003	0.013	0.002	0.22	1.45	0.006	0.052	0.005	0.12	1.18
10 yr	0.004	0.363	0.002	0.01	1.27	0.005	0.019	0.004	0.26	0.87
12.5 yr	-0.000	0.124	0.003	-0.00	0.95	0.005	0.036	0.005	0.14	0.81
15 yr	0.003	0.002	0.002	1.88	0.99	0.005	0.003	0.005	1.59	0.73
17.5 yr	-0.001	0.161	0.003	-0.01	0.21	0.004	0.103	0.005	0.04	0.62
20 yr	0.002	0.023	0.003	0.07	0.30	0.005	0.134	0.005	0.03	0.63
22.5 yr	0.002	0.023	0.003	0.07	0.26	0.005***	0.000	0.005	10.88	0.66
25 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
27.5 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
30 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.005$.

Table 9: Empirical Estimates: VC — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	0.473	18.860	0.187	0.03	3.91	1.732	11.028	0.585	0.16	2.40
0.1 yr	0.458	21.158	0.201	0.02	3.86	1.723***	0.237	0.575	7.26	2.40
2.5 yr	1.427	1.806	0.373	0.79	3.67	1.556	9.260	0.341	0.17	4.62
5 yr	1.475	2.296	0.326	0.64	4.44	1.454	8.251	0.323	0.18	4.77
10 yr	1.796	25.246	0.190	0.07	9.49	1.879	17.174	0.237	0.11	8.44
12.5 yr	1.970	4.329	0.205	0.46	9.57	2.002	10.569	0.159	0.19	13.06
15 yr	2.129	6.375	0.441	0.33	4.46	2.119	7.601	0.155	0.28	13.94
17.5 yr	2.224	5.140	0.243	0.43	9.01	2.164	7.186	0.195	0.30	11.27
20 yr	2.284	4.663	0.265	0.49	8.43	2.182	6.896	0.231	0.32	9.53
22.5 yr	2.304	4.669	0.266	0.49	8.41	2.184***	0.316	0.248	6.91	8.81
25 yr	2.303	4.669	0.270	0.49	8.29	2.179***	0.323	0.243	6.75	9.02
27.5 yr	2.315	4.607	0.271	0.50	8.31	2.189***	0.283	0.237	7.73	9.28
30 yr	2.326	4.542	0.268	0.51	8.46	2.193***	0.272	0.234	8.05	9.41
Panel B: Alpha (β_{Alpha})										
0 yr	0.020	0.115	0.009	0.17	1.61	-0.001	0.048	0.010	-0.03	0.45
0.1 yr	0.020	0.130	0.009	0.15	1.46	-0.001	0.001	0.010	-1.84	0.43
2.5 yr	0.007	0.009	0.005	0.81	1.35	0.001	0.043	0.003	0.03	0.60
5 yr	0.006	0.012	0.004	0.54	1.14	0.002	0.039	0.003	0.05	0.37
10 yr	-0.001	0.119	0.002	-0.01	-1.30	-0.004	0.095	0.001	-0.04	-4.27
12.5 yr	-0.004	0.006	0.001	-0.78	-5.86	-0.006	0.059	0.001	-0.11	-7.52
15 yr	-0.007	0.036	0.002	-0.18	-2.99	-0.008	0.048	0.001	-0.16	-7.27
17.5 yr	-0.007	0.032	0.001	-0.23	-9.95	-0.008	0.046	0.001	-0.18	-6.11
20 yr	-0.008	0.030	0.001	-0.27	-8.85	-0.008	0.045	0.002	-0.19	-5.30
22.5 yr	-0.008	0.030	0.001	-0.28	-8.18	-0.009***	0.001	0.002	-9.67	-4.97
25 yr	-0.008	0.030	0.001	-0.28	-8.14	-0.009***	0.001	0.002	-9.61	-5.07
27.5 yr	-0.008	0.029	0.001	-0.29	-8.45	-0.009***	0.001	0.002	-9.74	-5.22
30 yr	-0.009	0.029	0.001	-0.29	-8.66	-0.009***	0.001	0.002	-9.81	-5.36

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

short Horizons to 0.63–0.76 at long Horizons), showing that a second factor does not materially alter the market risk assessment. The asymptotic standard errors are erratic throughout, often exceeding 10 and occasionally reaching triple-digit values, which highlights the unreliability of asymptotic inference in this setting.

Venture Capital. Figure 15 displays qualitatively analogous but more extreme patterns, consistent with the higher idiosyncratic risk inherent to early-stage ventures. At short Horizons, the second-factor loadings are strikingly large: the IA loading reaches 5.1 under EW at the 0-year Horizon ($\mathrm{SR}_{CV} \approx 1.4$) and 4.2 under FW, the ROE loading starts at 1.5 (EW) and 1.0 (FW), the EG loading at approximately 3.5 (EW) and 6.6 (FW), while the ME factor is the exception with a subdued loading near zero or slightly positive (0.3 under EW, -0.03 under FW). All four second-factor loadings undergo dramatic sign reversals as the Horizon extends. The IA loading, for instance, turns negative beyond 10 years and eventually reaches -1.5 to -1.9 at long Horizons with large stability ratios (SR_{CV} exceeding -12). Similarly, EG and ROE produce large negative loadings at long Horizons ($\hat{\beta}_{\mathrm{EG}} \approx -0.9$, $\hat{\beta}_{\mathrm{ROE}} \approx -1.2$ under EW at 30 years). These sign reversals echo the behavior documented in the simulation study (Section 4.5), where two-factor models exhibited severe attenuation and sign changes in the second-factor loading across Horizons. The MKT loading in the VC two-factor models varies substantially across specifications: in the IA model it stabilizes around 1.4–1.5 at intermediate Horizons, while in the ROE and EG models it ranges from 0.4 to 2.0 depending on the Horizon and weighting scheme.

Summary. The q^5-factor results for BO and VC confirm the fragility of multi-factor SDF estimation documented in the preceding MKT + α analysis. While some second-factor loadings attain statistical significance at isolated Horizons (particularly ME for BO and short-Horizon IA for VC), none of the four non-market factors produces a stable, economically interpretable loading across the full Horizon spectrum. The pervasive sign reversals and the extreme sensitivity of point estimates to the compounding Horizon indicate that these results are driven by finite-sample noise rather than genuine systematic risk exposures. We conclude that the q^5-factor two-factor models do not improve upon the single-factor MKT benchmark for either asset class.

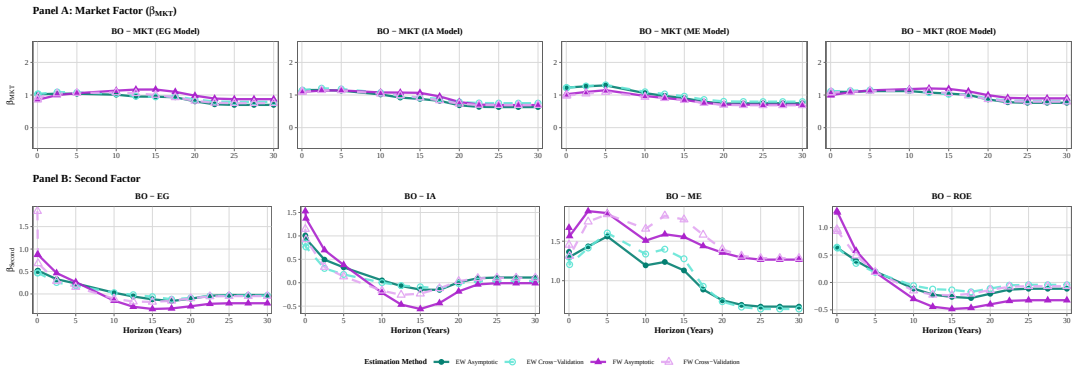

Figure 14: **Empirical results (q^5-factors):** Two-factor model for BO funds using VYP and maximum vintage year 2010.

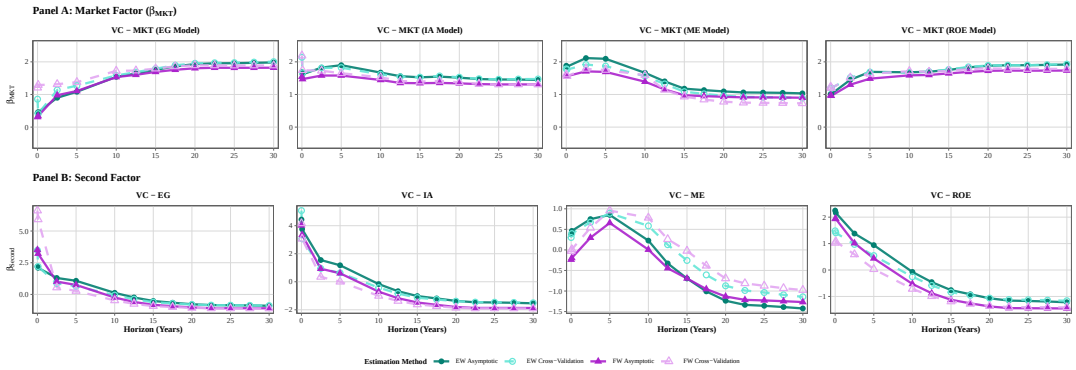

Figure 15: **Empirical results (q^5-factors):** Two-factor model for VC funds using VYP and maximum vintage year 2010.

5.4 Interpretation

The empirical results for Buyout and Venture Capital funds are best interpreted as evidence about what current private-equity cash-flow panels can identify under semiparametric SDF estimation, rather than as definitive evidence about the true factor structure of private equity.

Among the specifications considered, the single-factor MKT model is the most stable. For both BO and VC, the single-factor market model delivers economically plausible and comparatively stable estimates across Horizons and weighting schemes. For Buyout, the estimated market beta declines from approximately 1.27 at short Horizons to 0.67 at long Horizons under equal weighting. For Venture Capital, the corresponding range is 1.88–2.00 down to 1.04–1.13, consistent with the higher systematic risk exposure typically associated with early-stage investments. Most importantly, the cross-validation stability diagnostics remain comparatively strong across the Horizon grid, even when asymptotic standard errors

fluctuate erratically. In light of the simulation evidence and the limited cross-section of vintage-year portfolios, we interpret the asymptotic results in this application as a useful large-sample benchmark, but not as decisive finite-sample inference.

Adding α or a second factor materially weakens identification. Once a second parameter is introduced, the empirical estimates become substantially less stable. In the MKT + α specification, the intercept and market loading are only weakly separated in the available sample, so that the estimated α often absorbs variation that is attributed to MKT in the single-factor model. This produces implausible or near-collapsed market betas and highly erratic uncertainty measures. In the q^5-based two-factor models, the second-factor loadings vary sharply across Horizons and often attenuate toward, or pass through, zero at longer Horizons. We view these sign changes and sensitivity patterns as evidence of weak finite-sample identification in the present data environment, not as definitive evidence that the underlying economic factors are absent from private-equity payoffs.

The BO–VC split strengthens the identification interpretation. Analyzing Buyout and Venture Capital separately is informative because the two segments differ markedly in business models, leverage, cash-flow timing, and payoff profiles. Despite these economic differences, both sub-samples exhibit the same broad empirical pattern: relatively stable single-factor market estimates, destabilization once α is introduced, and fragile second-factor estimates. This common pattern across BO and VC suggests that the central difficulty is not merely the pooling of heterogeneous fund types. At the same time, it does not imply that BO and VC share the same true factor structure; rather, it indicates that the present sample does not support stable estimation of richer specifications for either segment.

The empirical patterns are qualitatively consistent with the simulation study. The empirical findings line up closely with the mechanisms highlighted in Section 4.5. First, the Horizon range at which the single-factor estimates appear most stable, approximately 15–20 years, is close to the 12.5–15-year region that minimizes MSE in the simulations. Second, the attenuation and occasional sign reversals in the empirical two-factor estimates resemble the weak-identification patterns observed in the simulated multi-factor experiments. Third, the gap between unstable asymptotic standard errors and more informative stability diagnostics is consistent with the simulation result that asymptotic approximations can substantially mischaracterize uncertainty in small samples. Taken together, the empirical results suggest that the finite-sample mechanisms isolated in the simulations are economically relevant in real-world private-equity cash-flow data.

Implications for empirical practice. For applied work, the main lesson is methodological. In the present sample, semiparametric SDF estimation for private-equity cash flows appears most credible when it is restricted to seasoned vintage-year portfolios, emphasizes parsimonious market-based specifications, and evaluates robustness over a Horizon grid. Richer specifications with α terms or additional factors may still be economically meaningful, but the current data do not identify

them with sufficient stability to support strong empirical conclusions. Accordingly, empirical claims in this setting should place greater weight on dependence-robust stability diagnostics, such as hv-block cross-validation, and less weight on formal asymptotic significance statements. A broader and more independent cross-section (through longer panels of seasoned funds, more observations within each vintage year, or both) would be needed before multi-factor semiparametric SDF estimates can be interpreted with comparable confidence.

6 Conclusion

Our generalized Least-Mean-Distance framework provides a flexible theoretical foundation for estimating Stochastic Discount Factor (SDF) models using non-traded cash flows. However, our Monte Carlo simulations reveal that semiparametric estimators in this setting face severe finite-sample challenges. Specifically, we demonstrate that the foundational Driessen et al. (2012) estimator exhibits substantially greater bias and variance than previously recognized, particularly concerning the estimation of abnormal performance (α). Our framework shows that this finite-sample bias can be dramatically mitigated by averaging pricing errors across multiple future dates spanning the fund lifecycle, while estimator variance is most effectively controlled by increasing the number of cross-sectional units (n).

In the data-sparse private-equity domain, where estimation relies on a limited cross-section of vintage-year portfolios, our empirical application shows that large-sample asymptotic approximations should be interpreted with caution. The asymptotic theory remains valuable as an organizing benchmark and as motivation for dependence-robust variance estimation, but in the empirical sample its implications are often dominated by finite-sample distortions. For that reason, we place greater weight on simulation evidence and on stability diagnostics tailored to overlapping cash-flow dependence (e.g., via hv-block cross-validation) than on formal asymptotic significance statements alone.

The empirical takeaway is therefore diagnostic rather than definitive. Current private-equity cash-flow panels appear informative about coarse market exposure in seasoned BO and VC portfolios, but substantially less informative about abnormal performance and richer factor decompositions. In our sample, single-factor MKT specifications are markedly more stable than models that add an α term or additional factors. We do not interpret this as evidence that private equity is truly priced by a single factor in population. Rather, we interpret it as evidence that, under currently available commercial cash-flow data and semiparametric pooled-cash-flow estimation, multi-parameter models are weakly identified and highly sensitive to finite-sample noise. Accordingly, parsimonious specifications should be preferred for empirical implementation unless materially broader cross-sectional information becomes available.

These results also bear on the literature on private-equity outperformance. Our findings do not imply that private-equity alpha is zero, nor do they imply that prior estimates of outperformance are necessarily incorrect. Rather, they indicate that semiparametric alpha estimates constructed from sparse pooled cash-flow panels can be weakly identified and highly sensitive to Horizon choice, sample construction, and

finite-sample noise. This qualification is directly relevant for interpreting prior evidence on private-equity outperformance, including Kaplan and Schoar (2005), Harris et al. (2014), Korteweg (2019, 2023), and related work. Differences across studies may reflect not only economic heterogeneity and dataset coverage, but also the limited ability of currently available cash-flow panels to separately identify abnormal performance from systematic risk exposure with high precision.

Expanding our SDF framework to robustly accommodate multiple risk factors requires imposing additional economic or statistical structure. Future research could further explore parametric assumptions for the cash flow generating process (Ang et al., 2018) or extract intra-period information from reported Net Asset Values (Brown et al., 2023). Methodologically, addressing the high estimator variance demands statistical techniques suited for sparse, high-dimensional settings. Approaches such as LASSO, ridge regression, or model averaging to regularize non-market coefficients, as well as ensemble learning methods that combine multiple weak learners, present promising paths forward.

Another vital avenue for future work involves explicitly modeling funding rounds, financial engineering, complex fee structures and carried interest within the data generating process. This would enable the estimation of company-level and gross-versus-net-of-fee SDFs, providing sharper evaluation models for both General Partners (GPs) and Limited Partners (LPs). Such structural refinements would naturally improve parameter accuracy and allow for a more rigorous analytical characterization of finite-sample bias. Additionally, carefully integrating intermediate NAVs as proxies for unobservable true valuations could help alleviate the timing risk premium bias inherent to terminal value concepts, though this requires downweighting actual cash distributions in favor of subjective accounting metrics.

Finally, we highlight the critical role of the compounding Horizon in empirical asset pricing for private markets. Reporting estimates on a future-value-based horizon grid yields an additional array of alternative parameter estimates that can be compared to the traditional NPV-based results of the Driessen et al. (2012) method. Our simulation study demonstrates that utilizing this future-value framework, i.e., averaging pricing errors over a broad set of discounting dates $\mathcal{T}$, yields vastly superior finite-sample properties compared to traditional net present value approaches. The simulation bias is minimized when the set of discounting dates corresponds approximately to the typical fund lifecycle. Developing a formal theoretical or empirical mechanism to endogenously determine the optimal configuration of $\mathcal{T}$ for a given fund sample and macroeconomic environment remains an important objective for future research.

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A Target shift and finite-sample stabilization under $\mathcal{T}$-averaging

This appendix clarifies why $\mathcal{T}$-averaging can improve finite-sample performance even though, for $H > 0$, it generally shifts the population target away from the structural parameter θ_0. The key mechanism is a tradeoff between: (i) a *population target shift* induced by post-investment timing terms through the Timing Risk Premium (TRP), and (ii) a *finite-sample stabilization effect* from averaging imperfectly correlated pricing errors across discounting dates.

Throughout this appendix, we specialize to the quadratic loss $L(x) = x^2$ and set $w_i = 1$, so that the population objective reduces to a nonlinear least-squares criterion. To keep the notation transparent, all moment calculations in this appendix are stated conditional on a fixed fund structure for a generic fund i (that is, conditional on J_i, the relevant investment dates, and the horizon set $\mathcal{T}_{i,H}$), while expectations continue to integrate over SDF and pricing-error randomness. The corresponding unconditional population statements follow by taking an outer expectation over the distribution of fund structures.

To keep the different objects distinct, we proceed in four steps. First, we define the horizon-specific population criterion and pseudo-true minimizer. Second, we identify the exact source of the averaged population moment shift at θ_0. Third, we show how that moment shift enters the gradient of the population criterion and thereby affects the pseudo-true parameter. Fourth, we present an exact MSE decomposition and then interpret the simulation evidence through a second-order finite-sample lens.

A.1 Setup and horizon-specific population target

For a given maximum compounding horizon H, let

$$\mathcal{T}_{i,H} := \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\},$$

where $T_i^{\max}$ denotes the largest discounting date available for fund i. Define the horizon-averaged pricing error by

$$\bar{\epsilon}_i^{(H)}(\theta) := \frac{1}{\mathrm{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta). \tag{27}$$

The associated population criterion is

$$Q_H(\theta) := \mathbb{E}\Big[\big(\bar{\epsilon}_i^{(H)}(\theta)\big)^2\Big], \tag{28}$$

and the horizon-specific pseudo-true parameter is defined by

$$\theta_H^\dagger := \arg\min_\theta Q_H(\theta), \tag{29}$$

whenever the minimizer exists and is unique. For notational simplicity, the conditioning on the fixed fund structure is suppressed in $Q_H(\theta)$ and in subsequent expectations. Thus, $\theta_H^\dagger$ is the target of the *population nonlinear least-squares criterion*. It is not defined as the zero of the averaged population moment, and those two objects need not coincide in general.

When $H = 0$, the estimator uses only the benchmark NPV discounting date $\tau = t_i^0$. Because no fund has cash flows before inception and every deal satisfies $t_i^0 \le t_{i,j}^{\text{Inv}}$, part (1) of Lemma 1 implies that, under correct specification,

$$\mathbb{E}\Big[\bar{\epsilon}_i^{(0)}(\theta_0)\Big] = 0.$$

If, in addition, the benchmark population criterion $Q_0(\theta)$ is uniquely minimized at θ_0, then

$$\theta_0^\dagger = \theta_0.$$

For $H > 0$, the inclusion of post-investment timing terms generally perturbs the population criterion, so that

$$\theta_H^\dagger \neq \theta_0 \qquad \text{in general.}$$

It is convenient to define the horizon-specific averaged population moment at the structural parameter, conditional on the fund structure of a generic fund i:

$$b_{H,i} := \mathbb{E}\Big[\bar{\epsilon}_i^{(H)}(\theta_0) \,\Big|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\text{Inv}}\}_{j=1}^{J_i}\Big]. \tag{30}$$

The next subsection shows that, under correct specification, $b_{H,i}$ is generated exactly by the post-investment TRP terms. The corresponding unconditional object is $\mathbb{E}[b_{H,i}]$.

A.2 Exact source of the averaged population moment at θ_0

Starting from the average pricing error in Equation (12) and the deal-level decomposition in Equation (3), we can write

$$\bar{\epsilon}_i^{(H)}(\theta_0) = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \epsilon_{\tau,i}(\theta_0) = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j}. \tag{31}$$

Taking expectations yields

$$b_{H,i} = \frac{1}{\text{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\text{Inv}}\}_{j=1}^{J_i}\right]. \tag{32}$$

For each (τ, j) term, use the identity

$$\mathbb{E}[XY] = \mathrm{Cov}(X, Y) + \mathbb{E}[X]\mathbb{E}[Y]$$

with

$$X = \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \qquad Y = \delta_{i,j}.$$

This gives

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\delta_{i,j}\right] = \mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right) + \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right]\mathbb{E}[\delta_{i,j}]. \tag{33}$$

Under correct specification, Equation (2) implies

$$\mathbb{E}[\delta_{i,j}] = \mathbb{E}\Big[\mathbb{E}[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}]\Big] = 0.$$

Hence only the covariance term remains.

Now consider $\tau \le t_{i,j}^{\mathrm{Inv}}$. Because $\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau$ is then $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$-measurable,

$$\begin{aligned}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\delta_{i,j}\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\mathbb{E}\Big[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\Big]\right] \\ &= 0.\end{aligned} \tag{34}$$

Since $\mathbb{E}[\delta_{i,j}] = 0$, Equation (34) also implies

$$\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right) = 0 \qquad \text{for all } \tau \le t_{i,j}^{\mathrm{Inv}}.$$

We therefore obtain the following exact characterization.

Lemma 2 (Exact source of the averaged population moment at θ_0)**.** *Under correct specification,*

$$b_{H,i} = \frac{1}{\mathrm{card}(\mathcal{T}_{i,H})} \sum_{\tau \in \mathcal{T}_{i,H}} \sum_{j=1}^{J_i} \mathbf{1}\{\tau > t_{i,j}^{\mathrm{Inv}}\}\, \mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \delta_{i,j} \,\middle|\, J_i, \mathcal{T}_{i,H}, \{t_{i,j}^{\mathrm{Inv}}\}_{j=1}^{J_i}\right). \tag{35}$$

In particular, benchmark NPV terms contribute zero to $b_{H,i}$, and the averaged population moment at θ_0 is shifted only by post-investment timing terms.

Proof. The result follows from Equation (32), the decomposition in Equation (33), the fact that $\mathbb{E}[\delta_{i,j}] = 0$ under correct specification, and the iterated-expectations argument in Equation (34) for all pairs (τ, j) satisfying $\tau \le t_{i,j}^{\mathrm{Inv}}$. □

Misspecification. Under model misspecification, $\mathbb{E}[\delta_{i,j}] \neq 0$, and the second term in Equation (33) no longer vanishes. In that case, the conditional averaged population moment at θ_0 contains both (i) the covariance term associated with the TRP and (ii) an additional mean component

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right]\mathbb{E}[\delta_{i,j}].$$

Without further sign or decay restrictions, horizon averaging need not eliminate that additional component. Accordingly, the clean correct-specification interpretation of $\mathcal{T}$-averaging as a tradeoff between a TRP-induced target shift and variance reduction no longer applies without qualification.

A.3 From averaged moment shift to pseudo-true target shift

Lemma 2 identifies the exact source of the conditional averaged population moment $b_{H,i}$ at θ_0. To connect that result to the parameter target, define

$$g_{i,H}(\theta) := \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta),$$

and assume that differentiation can be interchanged with expectation. To connect that result to the parameter target, define

$$g_{i,H}(\theta) := \nabla_\theta \bar{\epsilon}_i^{(H)}(\theta),$$

and assume that differentiation can be interchanged with expectation. Then

$$\nabla_\theta Q_H(\theta) = 2\,\mathbb{E}\Big[\bar{\epsilon}_i^{(H)}(\theta)\, g_{i,H}(\theta)\Big]. \tag{36}$$

Evaluating at θ_0 and decomposing the expectation around the conditional mean $b_{H,i}$ yields

$$\nabla_\theta Q_H(\theta_0) = 2\,\mathbb{E}[b_{H,i}\, g_{i,H}(\theta_0)] + 2\,\mathbb{E}\Big[\big(\bar{\epsilon}_i^{(H)}(\theta_0) - b_{H,i}\big) g_{i,H}(\theta_0)\Big]. \tag{37}$$

where for scalar X and vector Y, $\mathrm{Cov}(X,Y) := \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$.

Equation (37) makes two points precise. First, the TRP-driven conditional moment shift $b_{H,i}$ is one component of the criterion tilt at θ_0. Second, the pseudo-true target depends not only on that conditional mean shift, but also on the residual co-movement between the averaged pricing error and its parameter sensitivity.

Assume now that Q_H is twice continuously differentiable on an open convex neighborhood containing the line segment between θ_0 and $\theta_H^\dagger$, and that its Hessian is nonsingular throughout that set. Since $\theta_H^\dagger$ satisfies

$$\nabla_\theta Q_H(\theta_H^\dagger) = 0, \tag{38}$$

a first-order Taylor expansion around θ_0 gives

$$0 = \nabla_\theta Q_H(\theta_0) + \nabla_\theta^2 Q_H(\bar{\theta}_H)\,(\theta_H^\dagger - \theta_0), \tag{39}$$

for some intermediate point $\bar{\theta}_H$ between θ_0 and $\theta_H^\dagger$. Hence

$$\theta_H^\dagger - \theta_0 = -\big[\nabla_\theta^2 Q_H(\bar{\theta}_H)\big]^{-1} \nabla_\theta Q_H(\theta_0). \tag{40}$$

Combining Equations (37) and (40), the pseudo-true target shift is driven by the full gradient tilt of the population criterion at θ_0. Lemma 2 therefore identifies an exact channel through which the TRP moves the population target, but $b_{H,i}$ alone does not fully characterize $\theta_H^\dagger - \theta_0$ unless additional restrictions are imposed.

Remark 9 (A generic magnitude bound)**.** *For each post-investment term $\tau > t_{i,j}^{\mathrm{Inv}}$, Cauchy–Schwarz yields*

$$\left|\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}, \delta_{i,j}\right)\right| \leq \sigma\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right)\ \sigma(\delta_{i,j}).$$

Moreover, the averaging operator contributes the factor $1/\mathrm{card}(\mathcal{T}_{i,H})$ in Equation (35). Thus, large target shifts require either highly volatile SDF ratios, highly dispersed deal-level pricing errors, or strong alignment between these objects.

A.4 Exact MSE decomposition

Let $\hat{\theta}_H$ denote the sample extremum estimator associated with the sample analogue of Q_H. For any horizon H, the identity

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0) \tag{41}$$

holds exactly. Taking squared norms and expectations over the sampling distribution gives the following decomposition.

Proposition 7 (Exact MSE decomposition around the structural parameter)**.** *For any horizon H,*

$$\begin{aligned}\mathbb{E}\Big[\|\hat{\theta}_H - \theta_0\|^2\Big] &= \|\theta_H^\dagger - \theta_0\|^2 + 2(\theta_H^\dagger - \theta_0)^\top \mathbb{E}\Big[\hat{\theta}_H - \theta_H^\dagger\Big] \\ &\quad + \Big\|\mathbb{E}\Big[\hat{\theta}_H - \theta_H^\dagger\Big]\Big\|^2 + \mathrm{tr}\Big(\mathrm{Var}(\hat{\theta}_H)\Big).\end{aligned} \tag{42}$$

Proof. Write

$$\hat{\theta}_H - \theta_0 = (\hat{\theta}_H - \theta_H^\dagger) + (\theta_H^\dagger - \theta_0),$$

and expand the squared norm. Next, since $\theta_H^\dagger$ is nonrandom,

$$\mathrm{Var}(\hat{\theta}_H - \theta_H^\dagger) = \mathrm{Var}(\hat{\theta}_H),$$

so

$$\mathbb{E}\Big[\|\hat{\theta}_H - \theta_H^\dagger\|^2\Big] = \Big\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\Big\|^2 + \mathrm{tr}\Big(\mathrm{Var}(\hat{\theta}_H)\Big).$$ □

Proposition 7 separates four conceptually distinct objects:

1. the *population target shift* $\|\theta_H^\dagger - \theta_0\|^2$,
2. the *interaction term* $2(\theta_H^\dagger - \theta_0)^\top \mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]$, which links the target shift to the estimator's finite-sample bias around the pseudo-true parameter and can be either positive or negative,
3. the *squared finite-sample bias around the pseudo-true parameter* $\|\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger]\|^2$,
4. the *sampling variance* $\mathrm{tr}(\mathrm{Var}(\hat{\theta}_H))$.

This decomposition is exact. What remains model-dependent is how these terms move with H.

Heuristic link to second-order finite-sample bias. For nonlinear extremum estimators, standard higher-order expansions imply that

$$\mathbb{E}[\hat{\theta}_H - \theta_H^\dagger] = O(n^{-1}),$$

with a constant that depends on higher-order moments of the score and on local curvature; see, e.g., Nagar (1959); Rilstone et al. (1996); Newey and Smith (2004); Bao and Ullah (2007). In the present setting, those constants are naturally affected by the volatility and dependence structure of the objective contributions. This suggests the following heuristic channel: if $\mathcal{T}$-averaging reduces the volatility of $\bar{\epsilon}_i^{(H)}(\theta)$ without introducing a large target shift, it can improve finite-sample performance even when $H > 0$.

A convenient representation is

$$\mathrm{Var}\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = a_{i,H}^\top \Sigma_{i,H}(\theta) a_{i,H}, \tag{43}$$

where $a_{i,H}$ is the $\mathrm{card}(\mathcal{T}_{i,H}) \times 1$ averaging vector with entries $1/\mathrm{card}(\mathcal{T}_{i,H})$, and $\Sigma_{i,H}(\theta)$ is the covariance matrix of the vector

$$\big(\epsilon_{\tau,i}(\theta)\big)_{\tau \in \mathcal{T}_{i,H}}.$$

Remark 10 (A simple variance-reduction benchmark)**.** *Suppose, for illustration, that* $\mathrm{Var}(\epsilon_{\tau,i}(\theta)) = \sigma^2(\theta)$ *for all* $\tau \in \mathcal{T}_{i,H}$ *and that every pair of distinct dates has the same correlation* $\rho(\theta)$*. If* $m = \mathrm{card}(\mathcal{T}_{i,H})$ *and the implied equicorrelation covariance matrix is positive semidefinite, i.e.* $\rho(\theta) \geq -1/(m-1)$*, then*

$$\mathrm{Var}\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big) = \sigma^2(\theta)\left(\rho(\theta) + \frac{1-\rho(\theta)}{m}\right).$$

Whenever $\rho(\theta) < 1$*, this variance is strictly below* $\sigma^2(\theta)$*, and mild negative correlation lowers it even further. This special case illustrates the basic stabilization mechanism: with comparable marginal variances and less-than-perfect dependence, averaging lowers the variance of the objective contribution relative to a representative single-date moment.*

A.5 Interpretation and reconciliation with the simulation study

The simulation evidence in Section 4.5 can now be interpreted through Proposition 7.

At short horizons, the estimator relies on a small number of volatile pricing moments. This can make the nonlinear criterion unstable and can generate substantial finite-sample bias around the benchmark target. As H increases, averaging combines additional pricing moments that are correlated, but not perfectly so, and this can smooth the sample criterion and improve finite-sample behavior.

Against this benefit stands the population target shift characterized above. Once H is large enough to include post-investment timing terms, the criterion is generally no longer minimized at θ_0. The resulting TRP-induced shift can be economically

meaningful, but under the baseline calibration it is modest relative to the large finite-sample distortions observed at very short horizons.

This yields the central interpretation of the simulation results: for small to intermediate horizons, the stabilization benefit of $\mathcal{T}$-averaging can dominate the incremental target shift; for sufficiently long horizons, the marginal gain from additional averaging may weaken while the scope for target shift grows. Accordingly, the MSE-minimizing horizon need not be $H = 0$ and need not be arbitrarily large. Instead, it is generally *interior* and *DGP-dependent*.

Remark 11 (Why we do not impose a global monotonicity theorem)**.** *We do not claim that each MSE component is globally monotone in H without additional restrictions. In particular, the aggregate TRP contribution may contain offsetting covariance terms, and the effect of averaging on estimator variance depends on the covariance structure of the pricing moments and on the local geometry of the nonlinear criterion. For that reason, the appendix establishes the exact decomposition and the exact source of the averaged population moment shift, while the existence and location of the empirically relevant interior optimum are documented in the simulation study rather than asserted as a universal theorem.*

Remark 12 (Connection to GMM and misspecification)**.** *The role of $\mathcal{T}$-averaging is related to two familiar ideas. First, from a GMM perspective, averaging combines multiple imperfectly correlated pricing moments, which can improve finite-sample stability when those moments contain useful information. Second, from the perspective of misspecified moments, post-investment timing terms can be viewed as informative but potentially biased moment components: they perturb the population target, yet may still lower overall MSE when the induced target shift is small relative to the finite-sample stabilization they provide.*

Summary. The main message is not that the TRP "washes out" as H grows. Rather, $\mathcal{T}$-averaging trades off a population target shift against a potentially large reduction in finite-sample estimation error. Under the baseline DGP, the latter effect dominates over an economically relevant range of horizons, which is why the simulation study favors intermediate horizons rather than the NPV-only benchmark.

B Comparison to Driessen et al. (2012)

This section aligns our simulation and empirical designs with the original methodology of Driessen et al. (2012) to ensure direct comparability. Accordingly, we focus our analysis on estimates derived from NPV-based discounting (Horizon $H = 0$).

B.1 Comparison to simulation results

(Driessen et al., 2012, Table 1) report Monte Carlo results for $n = 50$ funds per vintage-year portfolio with an annualized idiosyncratic volatility of 108% ($\sigma_{\text{monthly}} = 0.3$). In our base case (S1), we use only $n = 20$ funds per vintage, which is more realistic for older vintages and smaller private capital asset classes beyond Buyout

and Venture Capital. Scenarios S2–S10 in Table 10 systematically vary the simulation design to identify which features drive the remaining discrepancy between our results and the Driessen et al. (2012) benchmark.

Several structural differences between the two simulation designs should be noted. First, Driessen et al. (2012) use a shifted lognormal distribution for the idiosyncratic error term (cf. their online appendix, Equation 9), which ensures that the error addend has a finite lower bound (approximately -1) and avoids the nonlinear distortions that arise when normally distributed errors occasionally produce extreme negative total returns. Second, they employ a "full Monte Carlo" approach and simulate the total market return R_{MKT} from a shifted lognormal distribution calibrated to S&P 500 moments (1980–2003), rather than conditioning on the historical path. Third, their deal investment timing follows a deterministic schedule (three deals per year, evenly spaced across the five-year investment period), rather than random uniform draws.

Interpretation of results. Under our original design with normal errors and $\sigma = 30\%$ (S3), the market beta is severely attenuated ($\hat{\beta} = 0.62$) and the spurious alpha (0.68%/month) is an order of magnitude larger than the benchmark (0.05%). Switching to the shifted lognormal error distribution (S4) modestly improves the beta estimate to 0.71 and reduces the alpha bias to 0.57%. Adding simulated market returns (S5) yields no further improvement ($\hat{\beta} = 0.66$), and deterministic investment timing (S7) produces a similarly attenuated beta of 0.70 without substantially reducing alpha variance.

The most effective modification is shortening the maximum deal holding period from 10 years to 5 years (S8), which significantly improves the beta estimate ($\hat{\beta} = 0.91$) and reduces the alpha bias to 0.30% without requiring lower idiosyncratic volatility. Combining this holding-period restriction with deterministic investment timing (S9) yields a similar beta estimate ($\hat{\beta} = 0.88$) and alpha bias (0.32%), with the deterministic timing contributing only a modest additional effect. By limiting the compounding duration, both scenarios restrict the geometric growth of idiosyncratic errors, reducing the frequency of degenerate alpha estimates and partially restoring the accuracy of the market beta. Tightening the alpha upper bound $\bar{\alpha}$ also dramatically affects results: at $\bar{\alpha} = 1\%$ (S6), 32% of draws hit the bound, and beta is close to the benchmark ($\hat{\beta} = 0.87$), but at the cost of artificially constraining the alpha search space. Interestingly, Driessen et al. (2012) report in their footnote 8 that they use a much higher upper limit for α: "In our application, putting an upper bound of 100% per year suffices to avoid this numerical issue." Combining all modifications, i.e., holding-period restrictions, deterministic timing, and a tighter alpha bound of $\bar{\alpha} = 1\%$ (S10), produces our closest approximation to the benchmark. Under this configuration, the beta estimate improves to 0.93 and the alpha bias falls to 0.27%, though 10% of draws still hit the alpha bound. However, even under this most favorable scenario (S10), our estimator still exhibits substantially more beta attenuation ($\hat{\beta} = 0.93$ vs. 0.98) and alpha bias (0.27% vs. 0.05%) than the Driessen et al. (2012) benchmark. This sensitivity reveals a fundamental tension in the two-factor model: the alpha term absorbs residual nonlinearity that the linear SDF cannot capture, and bounding it arbitrarily distorts the beta estimate.

Overall, the large remaining discrepancy between our simulation results and

those of Driessen et al. (2012), even after adopting their shifted lognormal errors and simulated market returns, implies that we were unable not reproduce the favorable small-sample properties reported in Driessen et al. (2012). Across all scenarios with $\sigma = 30\%$, the beta estimates range from 0.62 to 0.93, well below the Driessen et al. (2012) benchmark of 0.98, confirming the estimator's sensitivity to the interplay between idiosyncratic noise, compounding Horizon, and alpha bounds.

Table 10: Comparison to simulation (Driessen et al., 2012, Table 1)

	Market Beta (β)					Alpha (α, monthly %)					
Scenario	True	Mean	SD	25%	75%	True	Mean	SD	25%	75%	% Bound
(S1) $n = 20$, $\sigma = 20\%$, Normal	1.00	0.84	0.40	0.61	1.05	0.00	0.22	0.28	0.04	0.32	0%
(S2) $n = 50$, $\sigma = 20\%$, Normal	1.00	0.89	0.29	0.74	1.03	0.00	0.13	0.16	0.03	0.19	0%
(S3) $n = 50$, $\sigma = 30\%$, Normal	1.00	0.62	0.72	0.12	1.03	0.00	0.68	0.70	0.13	1.08	0%
(S4) $n = 50$, $\sigma = 30\%$, Shifted LN	1.00	0.71	0.72	0.23	1.12	0.00	0.57	0.67	0.07	0.91	0%
(S5) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT	1.00	0.66	0.79	0.11	1.06	0.00	0.71	0.79	0.03	1.34	0%
(S6) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, $\bar{\alpha} = 1\%$	1.00	0.87	0.69	0.41	1.24	0.00	0.48	0.50	0.06	1.00	32%
(S7) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Det. Timing	1.00	0.70	0.79	0.19	1.12	0.00	0.65	0.80	-0.01	1.20	0%
(S8) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5	1.00	0.91	0.54	0.64	1.19	0.00	0.30	0.52	-0.03	0.47	0%
(S9) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing	1.00	0.88	0.54	0.61	1.16	0.00	0.32	0.51	-0.01	0.53	0%
(S10) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing, $\bar{\alpha} = 1\%$	1.00	0.93	0.50	0.65	1.17	0.00	0.27	0.39	-0.01	0.52	10%
Driessen et al. 2012, Table 1											
Benchmark Estimates ($n = 50$, $\sigma_{\text{monthly}} = 30\%$)	1.00	0.98	0.32	0.80	1.15	0.00	0.05	0.2	-0.06	0.15	n/a

Notes: This table reports the small-sample properties of the generalized estimator for Horizon 0 (i.e., treating funds as held to maturity), mimicking the structure of Table 1 in Driessen et al. (2012). We simulate 1,000 Monte Carlo draws. All scenarios use 20 vintage years (1986–2005) with 15 deals per fund; unless stated otherwise, $n = 20$ funds per vintage, $\sigma = 0.2$ monthly idiosyncratic volatility, and a normal error distribution. The simulation dimensions vary as follows: n = number of funds per vintage-year portfolio; σ = monthly idiosyncratic standard deviation; *Normal* = Gaussian error term (default); *Shifted LN* = shifted lognormal error term (cf. Driessen et al. (2012)); *Sim. MKT* = total market return drawn from a shifted lognormal (S&P 500 1980–2003 moments) with a constant risk-free rate of 4% annualized, instead of conditioning on the historical path; *Det. Timing* = deterministic (evenly spaced) deal investment dates instead of random; *Max. Hold 5* = maximum holding period of a deal limited to 5 years (default 10 years); $\bar{\alpha}$ = upper bound on the alpha parameter in the optimization (default $\bar{\alpha} = 2\%$/month). The *% Bound* column reports the fraction of draws where the alpha estimate reached the upper optimization bound $\bar{\alpha}$. The final row presents the Benchmark Estimates obtained directly from Table 1 in Driessen et al. (2012) for comparative purposes. Alpha estimates are monthly and presented in percentages.

B.2 Comparison to empirical results

In this section, we evaluate the empirical stability of single- and multi-factor SDF specifications by estimating these models on our dataset (for the US-Dollar North America subsample with maximum vintage 2010) and comparing the results directly with the benchmark estimates published by Driessen et al. (2012). A key distinction is the underlying private market dataset: we use the Preqin cash flow database, which is also employed by Korteweg and Nagel (2016) and Ang et al. (2018), whereas Driessen et al. (2012) rely on the Thomson Venture Economics (TVE) fund-level dataset, which suffers from well-documented data quality issues such as forward-filling of stale KPIs (Harris et al., 2014). The cash flow summary of (Driessen et al., 2012, Table 3) indicates fund-value weighted vintage-year portfolios. (Driessen et al., 2012, Table 5) report results for Buyout (BO) and Venture Capital (VC) funds using both a two-factor model (S&P 500 MKT + α) and the

Fama and French (1993) four-factor model (MKT, SMB, HML, and α). In their two-factor specification, VC exhibits a market beta of 2.73 and a highly negative annualized alpha of -12.3%, while BO yields a market beta of 1.31 and an annualized alpha of -4.8%. These strongly negative alpha estimates can be at least partially attributed to the aforementioned data errors in TVE. Indeed, the BO alpha coefficient is statistically insignificant once bootstrap standard errors are taken into account, whereas the VC alpha remains significant. In their four-factor specification, the factor loadings are broadly plausible: BO exhibits $\hat{\beta}_{\text{MKT}} = 0.72$, $\hat{\beta}_{\text{SMB}} = 0.50$, and $\hat{\beta}_{\text{HML}} = 0.19$, while VC shows $\hat{\beta}_{\text{MKT}} = 2.20$, $\hat{\beta}_{\text{SMB}} = 1.83$, and $\hat{\beta}_{\text{HML}} = -0.48$. However, Driessen et al. (2012) note that the four-factor model does not significantly improve upon the two-factor specification and that the SMB and HML loadings are largely insignificant.

Table 11 and Table 12 present our empirical results for the single-factor (MKT), two-factor (α, MKT), three-factor (MKT, SMB, HML), and four-factor (α, MKT, SMB, HML) Fama and French (1993) models for BO, VC, and PE at Horizon 0, using the conservatively restricted sample with a maximum vintage-year cutoff of 2010.

We begin with the single-factor MKT model to establish a reliable baseline. Under this specification, market betas are consistently positive and theoretically plausible: the cross-validation fold averages ($\hat{\beta}_{\text{MKT}}$) are 1.23 for BO, 1.78 for VC, and 1.21 for PE under equal weighting (EW). The CV standard deviations verify that these estimates are highly stable across data subsets (e.g., CV standard deviations of 0.12 for BO and 0.09 for PE under EW). This result provides solid empirical backing that private equity returns exhibit strong, persistent co-movement with the broader market.

However, when explicitly allowing for an abnormal performance α intercept alongside the market factor, the estimation immediately collapses. In the two-factor model (α, MKT), the alpha parameter becomes highly sensitive to random, fund-specific noise compounded over a fund's lifetime. Even when restricting the sample to mature vintages, private equity datasets contain relatively few funds per year. As a result, there are simply not enough cross-sectional observations to adequately average out this idiosyncratic noise. Consequently, the optimizer overwhelmingly assigns inflated, positive alpha premiums (often hitting or nearing the 2% monthly bound, e.g., $\hat{\alpha} = 0.020$ for VC under EW and 0.008 for BO under FW), while the market betas are artificially driven toward zero (e.g., $\hat{\beta}_{\text{MKT}} = 0.14$ for BO and 0.18 for PE under FW). While asymptotic standard errors correspondingly explode or round to zero, the CV standard deviations reveal the actual parameter variation across data subsets. For instance, the cross-validation fold dispersion for the MKT estimate exceeds 0.6 for both BO and PE under FW, confirming that the collapsed point estimates are highly sensitive to the sample and untrustworthy.

This pathological behavior extends directly to the multi-factor models. When estimating the three-factor model (MKT, SMB, HML) *without* an alpha term, the estimates roughly mirror the MKT-only model (e.g., BO $\hat{\beta}_{\text{MKT}} = 1.35$ under EW), and the CV analysis shows reasonable out-of-sample stability, even though the asymptotic standard errors are vast. The SMB loading is consistently positive for BO and PE, but predominantly negative or near zero for VC. Notably, this sign for VC gen-

erally *reverses* the pattern reported by Driessen et al. (2012), who find a strongly positive VC SMB loading ($\hat{\beta}_{\text{SMB}} = 1.83$). The discrepancy likely reflects differences in the underlying datasets (Preqin vs. TVE) and sample periods, underscoring the severe fragility of multi-factor estimates in private equity.

But once the alpha term is reintroduced in the four-factor model (α, MKT, SMB, HML), the framework fails for the same reasons as the two-factor model. Because the alpha term absorbs the bulk of the remaining variance, the other factor loadings swing erratically: the asymptotic MKT coefficient for BO drops substantially (to 0.25 under FW), while SMB and HML loadings reach extreme values. The pervasive asymptotic standard errors of exactly (0.000) in these models often coincide with the estimates hitting the boundary constraints. The CV analysis exposes this reality: rather than being estimated with zero error, the parameters swing erratically across hold-out sets, with CV standard deviations for the factor loadings routinely exceeding 0.4, indicating that the multi-factor models are highly unreliable when alpha is included.

In conclusion, our empirical evaluation provides a strong, objective corroboration of the simulation-based warnings from Section 4.5: with the limited cross-section of only 20–40 vintage-year portfolios currently available in commercial private equity databases, any generalized SDF model attempting to simultaneously extract multiple pricing factors alongside an explicit alpha parameter is severely under-identified given the available cross-section. The inclusion of an unconstrained or loosely constrained abnormal performance term confounds the signal with cross-sectional noise, significantly destabilizing the factor model estimates. Therefore, until substantially more granular and independent cross-sectional data becomes cleanly available, we conclude that the single-factor market model remains the most robust and estimable specification for private equity SDF performance evaluation in our implementations.

Table 11: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Buyout and Venture Capital

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Buyout (BO)								
1-Factor (Asymp.)		1.268				1.197		
		(1.162)				(1.177)		
1-Factor (CV)		1.234				1.142		
		(0.119)				(0.137)		
2-Factor (Asymp.)	0.004	1.005			0.008	0.140		
	(0.006)	(1.070)			(0.189)	(25.676)		
2-Factor (CV)	0.006	0.588			0.008	0.122		
	(0.004)	(0.481)			(0.005)	(0.718)		
3-Factor (Asymp.)		1.389	1.296	0.415		1.252	1.870	0.296
		(1.588)	(4.991)	(0.988)		(2.471)	(6.876)	(2.989)
3-Factor (CV)		1.354	1.144	0.391		1.198	1.544	0.196
		(0.177)	(0.382)	(0.114)		(0.148)	(0.510)	(0.274)
4-Factor (Asymp.)	0.004	1.005	0.006	0.008	0.007	0.251	0.147	0.115
	(0.003)	(0.000)	(0.000)	(0.000)	(0.009)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.004	0.901	0.115	-0.005	0.007	0.420	0.548	-0.159
	(0.002)	(0.215)	(0.241)	(0.122)	(0.007)	(0.774)	(1.090)	(0.907)
Venture Capital (VC)								
1-Factor (Asymp.)		1.887				1.588		
		(1.768)				(1.655)		
1-Factor (CV)		1.775				1.592		
		(0.405)				(0.363)		
2-Factor (Asymp.)	0.020	0.467			-0.001	1.735		
	(0.121)	(19.892)			(0.050)	(11.554)		
2-Factor (CV)	0.014	0.732			0.004	1.409		
	(0.009)	(0.186)			(0.010)	(0.589)		
3-Factor (Asymp.)		2.013	-0.364	1.425		1.943	-0.444	1.395
		(0.755)	(2.207)	(0.440)		(0.856)	(1.689)	(1.521)
3-Factor (CV)		1.911	-0.330	1.209		1.887	0.051	1.112
		(0.453)	(0.430)	(0.946)		(0.666)	(0.720)	(1.444)
4-Factor (Asymp.)	0.020	1.384	0.680	4.096	0.001	1.620	-0.534	1.112
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.012	1.090	-0.171	1.534	0.005	1.780	-0.053	1.437
	(0.008)	(0.373)	(0.336)	(1.511)	(0.010)	(0.558)	(0.722)	(1.941)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

Table 12: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Private Equity

	Equal-Weighted (EW)				Fund-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Private Equity (PE)								
1-Factor (Asymp.)		1.216				1.134		
		(0.793)				(1.171)		
1-Factor (CV)		1.214				1.083		
		(0.087)				(0.136)		
2-Factor (Asymp.)	0.003	1.003			0.008	0.178		
	(0.001)	(0.801)			(0.595)	(88.860)		
2-Factor (CV)	0.004	0.912			0.007	0.154		
	(0.002)	(0.259)			(0.005)	(0.630)		
3-Factor (Asymp.)		1.252	0.790	0.471		1.177	1.695	0.240
		(13.846)	(4.351)	(10.009)		(4.248)	(4.996)	(2.624)
3-Factor (CV)		1.210	0.522	0.548		1.140	1.418	0.169
		(0.186)	(0.328)	(0.168)		(0.131)	(0.443)	(0.223)
4-Factor (Asymp.)	0.003	1.003	0.003	0.006	0.009	0.035	0.095	-0.177
	(0.001)	(0.000)	(0.000)	(0.000)	(0.008)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.004	0.924	0.035	0.033	0.006	0.367	0.343	-0.127
	(0.002)	(0.226)	(0.091)	(0.075)	(0.005)	(0.460)	(1.075)	(0.836)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) means across hold-out sets are reported. Standard errors (or standard deviations of the estimates for CV) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Fund-Size-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

C *Online Appendix:* Proofs for the horizon-specific asymptotic results

This subsection supplies detailed proofs for Proposition 2, Proposition 3, and Proposition 4. For notational convenience, fix a horizon $H \geq 0$ throughout and write

$$Z_i(\theta) := L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \qquad Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} Z_i(\theta).$$

Whenever convenient, re-index the observations by vintage year and within-vintage slot. That is, for vintage year $v = 1, \ldots, V$ and slot $\ell = 1, \ldots, n_v$, let $Z_{v,\ell}(\theta)$ denote the loss contribution of the ℓth unit in vintage year v. By Assumption 3(2), $n_v \leq \bar{n} < \infty$ uniformly in v. Hence the sample can be embedded in the increasing strip

$$\Lambda_V := \{(v, \ell) : v = 1, \ldots, V,\ \ell = 1, \ldots, \bar{n}\},$$

where empty slots are padded with zeros. Because $\bar{n}$ is fixed while $V \to \infty$, this embedding preserves the order of the sample size, and Assumption 3(2) implies

$$V \leq n = \sum_{v=1}^{V} n_v \leq \bar{n}V, \qquad \text{so that} \qquad n \asymp V.$$

Proof of Proposition 2. Let

$$\bar{Q}_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\big[Z_i(\theta)\big].$$

It suffices to show that

$$\sup_{\theta\in\Theta} \big|Q_{n,H}(\theta) - \bar{Q}_{n,H}(\theta)\big| \xrightarrow{p} 0, \qquad \sup_{\theta\in\Theta} \big|\bar{Q}_{n,H}(\theta) - Q_H(\theta)\big| \to 0,$$

because then the triangle inequality yields the claim.

We first verify the stochastic part. By Assumption 5(1), the array $\{\bar{\epsilon}_i^{(H)}(\theta) : i = 1, \ldots, n\}$ is uniformly NED on an α-mixing vintage-year process $\{\eta_v\}_{v\geq 1}$. Since $L(\cdot)$ is continuous and Lipschitz by Assumption 5(2), $Z_i(\theta) = L(\bar{\epsilon}_i^{(H)}(\theta))$ inherits the same NED structure, uniformly in $\theta \in \Theta$. Moreover, Assumption 5(3) gives a uniform L^p envelope for some $p > 2$, and Assumption 5(4) gives stochastic equicontinuity in θ. Re-indexing by (v, ℓ) therefore produces an array of random fields on the strip Λ_V satisfying the primitive conditions of Jenish and Prucha (2009, Theorem 6). Assumption 3(2)–(4) ensures that the field is observed over an increasing domain, not an infill domain: the width of the strip is fixed, dependence is indexed by the vintage-year distance, and bounded lifetimes prevent the effective dependence neighborhoods induced by overlapping cash-flow windows from expanding with n. Hence

$$\sup_{\theta\in\Theta} \left| \frac{1}{n}\sum_{i=1}^{n} \Big(Z_i(\theta) - \mathbb{E}[Z_i(\theta)] \Big) \right| \xrightarrow{p} 0.$$

Equivalently,

$$\sup_{\theta\in\Theta} \big|Q_{n,H}(\theta) - \bar{Q}_{n,H}(\theta)\big| \xrightarrow{p} 0.$$

It remains to pass from the sample-specific mean criterion $\bar{Q}_{n,H}(\theta)$ to the population target $Q_H(\theta)$ defined in Equation (16). By the ergodic homogeneity invoked immediately after Equation (16), the Cesàro average of the expectations converges to the generic-unit expectation as $V \to \infty$; because $n \asymp V$, the same holds under n-normalization. Therefore,

$$\sup_{\theta\in\Theta} \big|\bar{Q}_{n,H}(\theta) - Q_H(\theta)\big| \to 0.$$

Combining the last two displays proves

$$\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)| \xrightarrow{p} 0.$$

□

Proof of Proposition 3. We verify the hypotheses of the standard argmax theorem directly.

Step 1: continuity of the population criterion. For each unit i, the map $\theta \mapsto \bar{\epsilon}_i^{(H)}(\theta)$ is continuous on Θ because $\mathcal{T}_{i,H}$ is finite and the discount-factor ratios $\Psi_{\tau,t}(\theta)$ are continuous in θ. Since $L(\cdot)$ is continuous, $\theta \mapsto Z_i(\theta)$ is continuous as well. Assumption 5(3) supplies an integrable envelope uniformly in i and θ, so dominated convergence implies that

$$Q_H(\theta) = -\mathbb{E}\big[Z_i(\theta)\big]$$

is continuous on Θ. Because Θ is compact by Assumption 4, Q_H attains its maximum on Θ; by Assumption 6, this maximizer is unique and equals $\theta_H^\dagger$.

Step 2: uniform convergence. Proposition 2 yields

$$\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)| \xrightarrow{p} 0.$$

Step 3: argmax identification. Because $\hat{\theta}_H$ maximizes $Q_{n,H}$ over Θ, for every $\theta \in \Theta$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta).$$

In particular, with $\theta = \theta_H^\dagger$,

$$Q_{n,H}(\hat{\theta}_H) \geq Q_{n,H}(\theta_H^\dagger).$$

Subtracting and adding the population criterion gives

$$\begin{aligned} 0 &\leq Q_H(\theta_H^\dagger) - Q_H(\hat{\theta}_H) \\ &\leq \big|Q_H(\theta_H^\dagger) - Q_{n,H}(\theta_H^\dagger)\big| + \big|Q_{n,H}(\hat{\theta}_H) - Q_H(\hat{\theta}_H)\big| \\ &\leq 2\sup_{\theta\in\Theta} |Q_{n,H}(\theta) - Q_H(\theta)|, \end{aligned}$$

which converges to zero in probability. Hence

$$Q_H(\hat{\theta}_H) \xrightarrow{p} Q_H(\theta_H^\dagger).$$

Now let $\{n_k\}$ be any subsequence. By compactness of Θ, there exists a further subsequence, still denoted $\{n_k\}$ for notational simplicity, such that $\hat{\theta}_H \to \bar{\theta}$ almost surely along that subsequence for some random $\bar{\theta} \in \Theta$. By continuity of Q_H,

$$Q_H(\hat{\theta}_H) \to Q_H(\bar{\theta}) \qquad \text{almost surely along the subsequence.}$$

Since the same subsequence satisfies $Q_H(\hat{\theta}_H) \to Q_H(\theta_H^\dagger)$ in probability, it follows that $Q_H(\bar{\theta}) = Q_H(\theta_H^\dagger)$ almost surely. By uniqueness of the maximizer, $\bar{\theta} = \theta_H^\dagger$ almost surely. Therefore every convergent subsequence of $\hat{\theta}_H$ has the same limit $\theta_H^\dagger$, which proves

$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger.$$

This is exactly the conclusion of Newey and McFadden (1994, Theorem 2.1) specialized to the present criterion. □

Proof of Proposition 4. Define the sample score and sample Hessian by

$$S_{n,H}(\theta) := \nabla_\theta Q_{n,H}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} s_i^{(H)}(\theta), \qquad \mathcal{H}_{n,H}(\theta) := \nabla_{\theta\theta} Q_{n,H}(\theta).$$

Because $\theta_H^\dagger$ lies in the interior of Θ and $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$ by Proposition 3, $\hat{\theta}_H$ is an interior maximizer with probability approaching one. Hence the first-order condition holds with probability approaching one:

$$S_{n,H}(\hat{\theta}_H) = 0.$$

Applying a mean-value expansion of $S_{n,H}(\cdot)$ around $\theta_H^\dagger$ yields

$$0 = S_{n,H}(\theta_H^\dagger) + \mathcal{H}_{n,H}(\tilde{\theta}_H)\big(\hat{\theta}_H - \theta_H^\dagger\big),$$

where $\tilde{\theta}_H$ lies on the line segment joining $\hat{\theta}_H$ and $\theta_H^\dagger$. Rearranging gives

$$\sqrt{n}\,\big(\hat{\theta}_H - \theta_H^\dagger\big) = -\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1}\sqrt{n}\,S_{n,H}(\theta_H^\dagger). \tag{44}$$

We now analyze the two factors on the right-hand side.

Step 1: asymptotic distribution of the score. By definition,

$$\sqrt{n}\,S_{n,H}(\theta_H^\dagger) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n} s_i^{(H)}(\theta_H^\dagger).$$

Assumption 4 of Proposition 4 imposes a central limit theorem for these score contributions. In the present application, the required weak dependence is generated by the vintage-year random field, and the maintained NED and mixing conditions are exactly those covered by Jenish and Prucha (2012, Theorem 4) under increasing-domain asymptotics. Therefore,

$$\sqrt{n}\,S_{n,H}(\theta_H^\dagger) \xrightarrow{d} \mathcal{N}(0, \Lambda_H),$$

where

$$\Lambda_H = \lim_{n\to\infty} n \cdot \mathrm{Var}\Big[\nabla_\theta Q_{n,H}(\theta_H^\dagger)\Big].$$

Step 2: convergence of the sample Hessian. Because $\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger$, the intermediate point $\tilde{\theta}_H$ also satisfies $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$. Hence, with probability approaching one, $\tilde{\theta}_H \in \mathcal{N}_H$. Assumption 7 then gives

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) - \nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} 0.$$

Since Q_H is twice continuously differentiable in a neighborhood of $\theta_H^\dagger$, continuity of its Hessian and $\tilde{\theta}_H \xrightarrow{p} \theta_H^\dagger$ imply

$$\nabla_{\theta\theta} Q_H(\tilde{\theta}_H) \xrightarrow{p} \nabla_{\theta\theta} Q_H(\theta_H^\dagger) = \mathcal{H}_H.$$

Therefore,

$$\mathcal{H}_{n,H}(\tilde{\theta}_H) \xrightarrow{p} \mathcal{H}_H.$$

By assumption, $\mathcal{H}_H$ is nonsingular, so the continuous mapping theorem yields

$$\mathcal{H}_{n,H}(\tilde{\theta}_H)^{-1} \xrightarrow{p} \mathcal{H}_H^{-1}.$$

Step 3: conclusion. Substituting the previous two steps into Equation (44) and applying Slutsky's theorem gives

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}\big(0,\, \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}\big).$$

This is the extremum-estimator central limit theorem of Newey and McFadden (1994, Theorem 3.1), with the non-iid score CLT supplied by the increasing-domain NED random-field result of Jenish and Prucha (2012, Theorem 4). □